VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Nguyen Duc Khanh

ENHANCING OBJECT DETECTION PERFORMANCE UNDER FOGGY CONDITIONS THROUGH FUSION DOMAIN ADAPTATION

BACHELOR'S THESIS

Major: Computer Science

HANOI - 2026

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Co-Supervisor: M.Sc. Pham Duc Anh

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Declaration

My name is Nguyen Duc Khanh, a student of class K67I-CS2, Computer Science program. I hereby declare that my graduation thesis, entitled "Enhancing Object Detection Performance under Foggy Conditions through Fusion Domain Adaptation", is the result of my own research and has not been submitted for the award of any degree at any higher-education institution, including the VNU University of Engineering and Technology. I affirm that, throughout the development of this work, I have not copied content from any source, nor have I used the research results of any individual or organization without clear citation. This work was conducted under the supervision of Dr. Nguyen Duc Anh and M.Sc. Pham Duc Anh and contains no source code copied or modified from any individual or organization without proper authorization. In addition, parts of the source-code refactoring and documentation process were assisted by Anthropic's Claude AI. All data, results, and references used in this thesis are accurate, truthful, and fully cited in accordance with the regulations. Should any violation of the commitments above be found, I take full responsibility under the regulations of the VNU University of Engineering and Technology.

Hanoi, 21 May 2026

Student

Nguyen Duc Khanh

Abstract

Object detection under foggy weather is an important problem in computer vision, playing a critical role in outdoor systems such as autonomous driving and intelligent surveillance. However, modern detectors such as YOLO and Faster R-CNN, despite achieving high accuracy under normal weather conditions, suffer significant performance degradation when deployed in foggy environments. This degradation stems from atmospheric light scattering, contrast reduction, and the distribution shift between the labeled training domain and the unlabeled deployment domain. To address this challenge, this thesis proposes **FusionDA**, a semi-supervised domain adaptation method integrated into an object detection model, combining three complementary mechanisms. First, generating cross-domain image pairs to bridge the domain gap at the image input level. Second, applying a Mean Teacher framework with knowledge distillation to exploit unlabeled target-domain data through high-quality pseudo-labels. Third, incorporating a Gradient Reversal Layer module together with a consistency loss to explicitly enforce the backbone to learn domain-invariant features. Experiments on the standard Cityscapes and Foggy Cityscapes benchmark show that FusionDA achieves the highest HM_{50} of 55.37, surpassing the strongest existing DAOD baseline with HM_{50} = 46.14) improvements by +20.0%. Notably, on the RTTS natural-fog test set entirely unseen during training, FusionDA improves mAP_{50} by +77.0% relative over the baseline - a relative improvement that substantially exceeds the gain on Foggy Cityscapes (+28.6%), confirming that the method learns fog-invariant representations rather than overfitting to the synthetic distribution. Additional analyses based on Maximum Mean Discrepancy, UMAP visualization, and C2PSA attention maps further confirm that FusionDA genuinely narrows the domain gap at the feature representation level and avoids the catastrophic forgetting commonly observed in DAOD methods. The full source code, checkpoints, and generated datasets are made publicly available to ensure reproducibility.

Keywords: Object detection, Domain adaptation, Semi-supervised learning, Foggy weather, FusionDA, YOLO26.

Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Full Term	Meaning / Description
ALDI	Align and Distill	A DAOD framework that combines feature alignment with self-distillation
AODNet	All-in-One Dehazing Network	An end-to-end deep dehazing network that directly predicts a clean image from a foggy input
AP	Average Precision	Average precision - the area under the precision-recall curve
BCE	Binary Cross-Entropy	Binary cross-entropy loss, used for two-class classification
CNN	Convolutional Neural Network	A convolutional neural network
CUT	Contrastive Unpaired Translation	An unpaired image translation model based on contrastive learning
CycleGAN	Cycle-Consistent Generative Adversarial Network	An unpaired image-to-image translation GAN based on the cycle-consistency constraint
DA	Domain Adaptation	A technique that enables a model to perform well on a domain different from the training domain
DANN	Domain Adversarial Neural Network	A domain-adversarial neural network
DAOD	Domain Adaptive Object Detection	Object detection with domain adaptation
EMA	Exponential Moving Average	Exponential moving average - the update rule for the teacher model's weights

FN	False Negative	False negatives - ground-truth objects missed by the detector
FP	False Positive	False positives - regions containing no object but incorrectly detected
FPN	Feature Pyramid Network	A feature pyramid network for multi-scale feature extraction
GAP	Global Average Pooling	Global average pooling over the feature map
GRL	Gradient Reversal Layer	The gradient reversal layer, used to learn domain-invariant features
HM	Harmonic Mean	The harmonic mean of the performance on the two domains
IoU	Intersection over Union	The intersection-over-union ratio between two regions, measuring the overlap between a predicted bounding box and its ground truth
LN	Layer Normalization	Layer normalization, which stabilizes training
mAP	Mean Average Precision	The mean of AP across all classes - the primary metric for object detection
MLP	Multi-Layer Perceptron	A multi-layer feed-forward neural network
MMD	Maximum Mean Discrepancy	A distance measure between two distributions in feature space
MT	Mean Teacher	Mean teacher - a semi-supervised learning framework based on a student-teacher pair
R-CNN	Region-based Convolutional Neural Network	A convolutional network based on region proposals
TP	True Positive	True positives - ground-truth objects correctly detected

UMAP	Uniform Manifold Approximation and Projection	A nonlinear dimensionality-reduction technique for data visualization
YOLO	You Only Look Once	A family of real-time, single-stage object detectors

Chapter 1

Introduction

Outdoor object detection under adverse weather, particularly fog, is a fundamental problem in computer vision and critically affects the safety of autonomous driving and intelligent surveillance systems. While modern detectors such as the YOLO family [1] and Faster R-CNN [2] achieve high accuracy in the clear domain, their performance degrades severely when deployed in the foggy domain due to light scattering and reduced contrast [3, 4]. An illustration is shown in Figure 1.1.

To address this problem, the research community has explored three representative approaches. The first is (1) restoring the image via *image dehazing* before passing it to the detector, with representative architectures including AOD-Net [5], FFA-Net [6], DehazeFormer [7], and C2PNet [8]. The second is (2) training the detector on *synthetic foggy data* generated from the atmospheric scattering model or physics-based simulation [9–12]. Finally, the most widely adopted approach is (3) *domain adaptation*, in which the model learns to narrow the distributional gap between the clear and foggy domains through feature alignment, adversarial learning, or self-distillation [13–16].

However, each approach has its own limitations. Approach (1) typically optimizes perceptual image quality rather than detection performance, causing loss of critical semantic information in dense-fog regions. Approach (2) substantially improves performance on the foggy domain but degrades it under normal conditions, indicating that the model fails to capture the complex generality of real-world environments. Approach (3) directly exploits unlabeled foggy data and is therefore the most promising; however, representative adversarial feature-alignment methods such as DA-Detect [4], or Mean-Teacher-based self-distillation methods such as ALDI [16], typically narrow the domain gap only at the feature level, ignore the distribution shift at the input-image level, and fail to systematically exploit the physical structure of fog. In particular, these representative solutions share a core issue: **the model does not learn an invariant representation of fog and remains biased toward the training domain** when relying solely on conventional methods. This motivates the approach of this thesis: combining the strengths of (2) and (3) by actively generating cross-domain image pairs based on the atmospheric scattering model and a dehazing network, while integrating a semi-supervised domain-adaptation mechanism to fully leverage the foggy domain data.

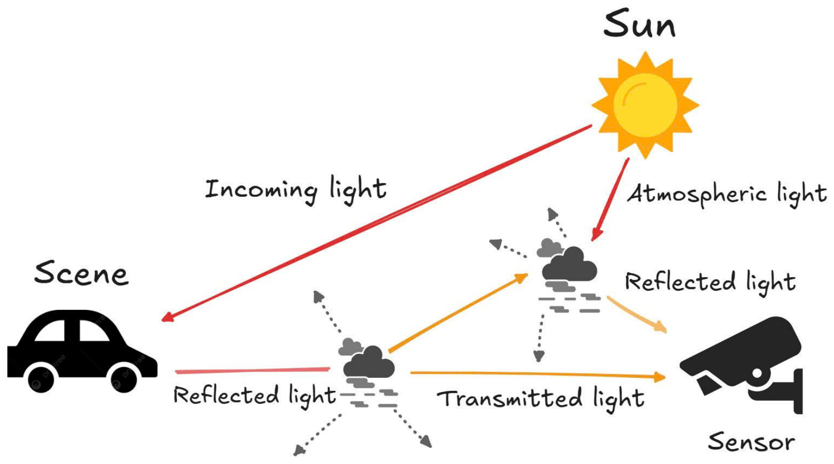

Figure 1.1: The atmospheric scattering model. Light from the source illuminates the scene; part of it is reflected, while another part is scattered by fog particles and absorbed before reaching the sensor. In addition, ambient light produced by multiple scattering reaches the sensor directly, brightening the captured image and reducing its contrast.

This is the fundamental challenge that severely degrades model performance when transferring from training data to real-world environments. The pronounced differences in image style, illumination, and image quality between the clear and foggy domains create a substantial domain gap [17]. Conventional machine-learning methods typically assume that training data follow an identical distribution, and consequently fail to generalize invariant features under weather and environmental changes. Many recent studies attempt to mitigate this domain bias by aligning features at different levels, ranging from the global image level [13] to the local-region level [14] and the instance level [18, 19]. However, these methods operate primarily on intermediate feature spaces, whereas the discrepancy between the foggy and clear domains can already emerge at the input-image level due to scattering and contrast reduction. When the image-level gap is left unclosed, domain-style noise propagates into deeper feature layers, undermining alignment at later stages [20, 21]. In addition, adversarial learning based on the Gradient Reversal Layer (GRL) [22] often suffers from optimization instability due to the minimax objective between the feature extractor and the domain classifier, particularly when applied directly to single-stage architectures [23]. These two limitations suggest that an effective domain-adaptation mechanism must simultaneously (i) actively close the domain gap at the input-image level before feature extraction, and (ii) stabilize adversarial training when employing GRL.

To address the above limitations, this thesis proposes **FusionDA**, a semi-supervised domain adaptation mechanism integrated into a single-stage object detection architecture. The core idea of FusionDA is to combine image-style translation, which narrows the domain gap at the image level, with knowledge distillation, which exploits information from unlabeled foggy data. The input consists of a fully labeled clear-weather street-scene dataset and a completely unlabeled foggy street-scene dataset, together with additional sets derived from these two. The output is an object detector that performs in a balanced manner on both the clear and foggy domains.

FusionDA proceeds in two main phases. In phase 1, FusionDA employs a depth-prediction model together with the atmospheric scattering equation and a dehazing network to synthesize new data for both domains. This phase aligns the style distribution between the two domains and directly resolves the data-level domain shift at the input. In phase 2, FusionDA applies the Mean Teacher framework [24] to produce high-quality pseudo-labels on the foggy data. This allows the model to learn salient object features without manual annotations. In parallel, a feature-consistency loss is enforced between the clear and pseudo-foggy image pairs, compelling the backbone to extract representations independent of domain style. Finally, the method integrates a Gradient Reversal Layer (GRL) module to strengthen domain invariance at the global feature level, thereby improving generalization across both domains.

To demonstrate the effectiveness of the proposed method, we conduct experiments on the standard Cityscapes [25] (clear domain) and Foggy Cityscapes [9] (synthetic foggy domain) benchmark, using multiple model variants including YOLO26 [26] for training and evaluation. To further verify generalization under real-world deployment, we extend the evaluation to a zero-shot setting on the natural-fog test sets RTTS [27], FoggyDriving [9], and FoggyZurich [28]. The baselines include conventional training and representative domain-adaptation methods (DA-Detect, ALDI++). The experimental results show a clear improvement of the proposed method over the baselines. Specifically, on the harmonic-mean metric $\mathrm{HM}_{50}^{(3)}$ across the three domains (clear, synthetic fog, natural fog), the strongest existing DAOD baseline reaches only 46.14 (DA-Detect R50-FPN), whereas the proposed method achieves 54.22 on YOLO26-s and peaks at 55.37 on YOLO26-l (a relative improvement of +17.5% to +20.0%). More notably, on the zero-shot RTTS set, FusionDA improves mAP_{50} by +24.87 absolute points (+77.0% relative) over the baseline

- a margin that substantially exceeds the gain on Foggy Cityscapes (+28.6%), confirming that the method genuinely learns fog-invariant representations rather than overfitting to the synthetic distribution. In addition, a supplementary experiment on face detection (WIDERFACE-easy [29]) with synthetic fog corruption from the `imagecorruptions` library [30] shows that FusionDA's improvement trend - particularly pronounced on small objects - is preserved when applied to single-class detection under a different fog corruption, reinforcing the generality of the proposed method.

In summary, the main contributions of this thesis are as follows:

- Propose FusionDA, a semi-supervised domain-adaptation framework for single-stage object detectors that combines the Mean Teacher framework with dedicated functional modules and a Gradient Reversal Layer module to narrow the domain gap simultaneously at the image and feature levels.
- Conduct a comprehensive evaluation of the proposed method on the standard Cityscapes and Foggy Cityscapes benchmark, together with zero-shot evaluation on real-world foggy sets, comparing against representative DAOD methods (DA-Detect, ALDI++) on both one-stage and two-stage architectures, along with a detailed ablation study that quantifies the contribution of each component, and a supplementary experiment on single-class detection (WIDERFACE-easy) under synthetic fog corruption to verify generality.
- Release the source code[1], model checkpoints[2], and the generated datasets publicly to support the research community and ensure reproducibility of all experimental results.

The remainder of this thesis is organized as follows. Chapter 2 presents the background, covering the fundamental concepts and the problem definition. Chapter 3 describes the proposed method in detail. Chapter 4 reports the experimental setup, comparative results, and discussion. Finally, we conclude and outline future directions.

[1] `https://github.com/khasnhmissu/FusionDA`

[2] `https://byvn.net/VKY1`

Chapter 2

Background

2.1 Object detection

Object detection has a long history of development that has produced a wide variety of methods. Classical works [31, 32] typically formulate object detection as sliding-window classification. The rise of deep convolutional neural networks (CNNs) [33] originated in object detection, where the early successes of CNNs triggered a major methodological shift.

Object detection has two main lines: two-stage and one-stage. Two-stage models (e.g., R-CNN [34] and Fast/Faster R-CNN [2, 35]) first generate region proposals and then classify them; this yields high accuracy, particularly on small objects, at the cost of low speed and a complex pipeline. In contrast, one-stage models (e.g., YOLO [1], SSD [36], and RetinaNet [37]) cast detection as a single regression problem, predicting bounding boxes and class labels in a single forward pass. Their main advantage is high speed, which makes them well suited to real-time systems.

For example, YOLO has demonstrated that it can sustain tens of FPS while substantially surpassing the AP of contemporary real-time detectors [1]. While RetinaNet introduces the Focal Loss objective to balance classes and improve accuracy, YOLO remains superior in speed. Within this scope, we select YOLO as the primary analysis target because of its real-time orientation, particularly with the most recent variants that achieve a strong balance between speed and accuracy.

A typical object detector consists of three components: a backbone, a neck, and a head. The backbone, built from convolutional layers, extracts and encodes features from the input image. The backbone is typically pre-trained on large image datasets. The neck sits between the backbone and the head, and is responsible for learning multi-scale features through architectures such as the Feature Pyramid Network [38], the Path Aggregation Network [39], and others. The head receives features from the neck and predicts bounding boxes together with the probabilities over c classes. The output of an object detector is a vector whose components are described in Equation 2.1.

$$y^T = [p_0, \underbrace{\langle t_x, t_y, t_w, t_h \rangle}_{\text{bounding box}}, \quad \underbrace{\langle p_1, p_2, ..., p_c \rangle}_{\text{predicted class probabilities}} \quad]. \tag{2.1}$$

where y is the detector's output vector, c is the number of classes, p_0 is the predicted objectness probability that the bounding box contains an object, t_x and t_y are the coordinates of the box center, t_w and t_h are the box width and height, and $p_1, p_2, \ldots, p_c$ form the vector of predicted probabilities over the c classes.

2.2 Domain adaptation

A fundamental assumption in conventional machine-learning models is that the training and test data are drawn from the same statistical distribution. This allows the model to learn a prediction function h with strong generalization. The problem is formulated as the minimization of the expected risk in Equation 2.2.

$$R(h) = \mathbb{E}_{(x,y)\sim P(x,y)}[\mathbf{L}(h(x), y)]. \tag{2.2}$$

where $\mathbf{L}$ is the loss function measuring the discrepancy between the prediction and the ground-truth label. However, in many real-world applications such as computer vision, signal processing, and behavior analysis, data conditions continually change with the environment, the device, the time, or the style. In such cases, the distribution of the new data no longer matches the distribution used to train the model. This phenomenon is known as *domain shift* [40].

Each data domain is described by a triple $D = \{\mathbf{X}, \mathbf{Y}, P(x, y)\}$, where $\mathbf{X}$ is the feature space, $\mathbf{Y}$ is the label space, and $P(x, y)$ is the joint distribution. Given a source domain S and a target domain T, domain shift occurs when $P_S(x, y) \neq P_T(x, y)$. In this case, a model trained on the source domain may perform poorly on the target domain. The goal of *domain adaptation* is to learn a model from source-domain data that remains effective on the target domain, even when target-domain labels are scarce or unavailable. The expected risk on the target domain can be rewritten in terms of the source distribution as in Equation 2.3.

$$R_T(h) = \mathbb{E}_{(x,y)\sim P_S(x,y)}\left[\frac{P_T(x, y)}{P_S(x, y)}\mathbf{L}(h(x), y)\right]. \tag{2.3}$$

The ratio between the two distributions acts as a correction factor that reduces the discrepancy between the domains. However, fully estimating $P_T(x, y)$ is typically infeasible due to the lack of labels in the target domain.

A representative model of domain adaptation is the Domain Adversarial Neural Network (DANN). It consists of a feature extractor G_f, a label predictor G_y, and a domain discriminator G_d. The optimization objective is given in Equation 2.4.

$$\min_{\theta_f, \theta_y} \max_{\theta_d} \left(\mathbf{L}_y(G_y(G_f(x)), y) - \lambda \mathbf{L}_d(G_d(G_f(x)), d) \right). \tag{2.4}$$

where $\mathbf{L}_y$ is the label-classification loss and $\mathbf{L}_d$ is the domain-discrimination loss. The adversarial optimization forces G_f to learn features that are simultaneously discriminative for the main task and indistinguishable across domains, thereby reducing domain shift.

In summary, domain adaptation provides a principled approach to handling distribution shifts in real-world systems by learning domain-invariant features. Modern methods, especially when combined with deep learning, continue to expand the generalization capability of models across diverse data environments.

2.3 Knowledge distillation framework

Knowledge distillation is a technique that transfers knowledge from a large, high-accuracy model (the *teacher*) to a smaller, more compact model (the *student*) [41]. The core idea is that, instead of learning only from binary correct/incorrect labels (hard labels), the student learns from the predictive outputs of the teacher.

This output is not a single class but a probability distribution over all classes, indicating how strongly the teacher *believes* in each class. This distribution carries more information than the hard label because it reflects the teacher's view of inter-class similarities (for instance, a cat may be more similar to a dog than to a car). To make these inter-class relationships more explicit, the model's output is "softened" through a parameter called the *temperature* T: a larger T yields a softer distribution that is less concentrated on a single class.

Specifically, given the student logits z_i and the teacher logits v_i, the softened probability distributions are computed as in Equation 2.5.

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \qquad p_i = \frac{\exp(v_i/T)}{\sum_j \exp(v_j/T)}. \tag{2.5}$$

where $i \in \{1, 2, \ldots, C\}$ indexes the current class and C is the total number of classes; z_i and v_i are the student and teacher logits for class i, respectively; j ranges over all C classes in the normalization denominator; $T \geq 1$ is the temperature parameter controlling the softness of the distribution; and q_i and p_i are the softened probabilities of the student and the teacher for class i, respectively.

The student is trained jointly with two objectives: (i) mimicking the teacher's softened distribution at temperature T, and (ii) predicting the true label at the standard temperature $T = 1$. The combined loss is given in Equation 2.6.

$$\mathbf{L} = \alpha\, T^2\, H(p_T, q_T) + \beta\, H(y, q_{T=1}). \tag{2.6}$$

where $H(\cdot, \cdot)$ denotes the cross-entropy between two probability distributions; $p_T = (p_1, \ldots, p_C)$ and $q_T = (q_1, \ldots, q_C)$ are the softened teacher and student distributions at temperature T; $q_{T=1}$ is the student's distribution at the standard temperature $T = 1$ (equivalent to the ordinary softmax); y is the ground-truth label represented as a one-hot vector; α and β are hyperparameters balancing the distillation and supervised classification objectives; and the T^2 factor in front of the distillation term compensates for the $1/T^2$ gradient attenuation induced by higher temperatures, ensuring that the two objectives have comparable magnitudes during optimization.

An important variant of knowledge distillation in semi-supervised learning is the *Mean Teacher* framework [24], in which the teacher is not a separately pre-trained network but an exponential moving average (EMA) of the student itself. The stability of this slowly updated teacher enables reliable pseudo-label generation on unlabeled data, forming the foundation of many modern DAOD methods such as ALDI [16] and serving as a core component of FusionDA (Chapter 3).

In summary, learning from the teacher's softened distribution plays a crucial role: even when data are scarce or unlabeled, the student receives fine-grained guidance from the teacher on how to discriminate between classes. As a result, a small model can approach the performance of a large model with substantially lower computational cost, making it well suited to real-world deployment.

2.4 Standard fine-tuning methods

With the growing availability of labeled data and the continual improvement of deep learning models, *fine-tuning* has become the default choice in many object-detection systems.

Rather than training from scratch, reusing weights from a pre-trained model shortens training time, reduces resource requirements, and improves initial performance. For modern networks such as YOLO26 in particular, a well-designed fine-tuning strategy can yield strong performance even in distinct data environments.

Fine-tuning in object detection typically starts from a model pre-trained on a large dataset (e.g., COCO) and further adjusts it to the new dataset. For YOLO, the pre-trained backbone is commonly reused for generic feature extraction, while only the head is retrained to recognize the new object classes. According to the YOLO documentation, freezing the early layers and updating only the parameters of the final layers substantially reduces training time and resource usage, although it may slightly lower the final accuracy. The standard YOLO architecture consists of a backbone and a head. During fine-tuning, the backbone parameters are typically kept frozen, and only the head is retrained to adapt to the new class set or domain.

2.5 Domain adaptation for object detection (DAOD)

When object-detection systems are deployed in real-world environments, they often encounter the data-level domain shift problem, where the deployment data differ substantially from the training data. When training and test data differ significantly, domain-adaptation methods are employed to mitigate the resulting performance drop. In this section, we introduce two prominent methods used as baselines for comparison with the proposed approach; both have been validated for effectiveness and are widely cited in the research community.

A general approach is the Domain Adversarial Neural Network (DANN), proposed by Ganin et al. [22]. DANN augments a network with a domain classifier and a gradient reversal layer, training the model jointly to recognize objects and to adversarially produce features that are difficult to distinguish across domains. The resulting features are simultaneously discriminative for the main task and invariant to inter-domain differences. This approach has proven effective for classification and extends naturally to object detection, where it improves generalization across environments. The Gradient Reversal Layer (GRL) is the core component of DANN [22], placed between the feature extractor G_f and the domain discriminator G_d. Its goal is to learn a feature space $\mathbf{z} = G_f(\mathbf{x})$ that simultaneously satisfies two properties. First, *discriminativeness*: the features retain enough information to accurately classify object labels on the source domain. Second, *domain* *invariance*: the distributions $p(\mathbf{z} \mid \mathcal{D}_s)$ and $p(\mathbf{z} \mid \mathcal{D}_t)$ cannot be distinguished by any domain classifier.

Forward pass. The GRL acts as an identity transformation, passing the input features through unchanged, as in Equation 2.7.

$$R_\lambda(\mathbf{x}) = \mathbf{x}. \tag{2.7}$$

Backward pass. During backpropagation, the GRL multiplies the gradient by a negative constant $-\lambda$ before passing it back to the preceding layers, as in Equation 2.8.

$$\frac{\partial R_\lambda}{\partial \mathbf{x}} = -\lambda \mathbf{I}. \tag{2.8}$$

where $\mathbf{I}$ is the identity matrix. This reversal mechanism is equivalent to performing *gradient ascent* (loss maximization) on G_d while performing *gradient descent* (loss minimization) on G_f, forming a unified minimax problem.

Domain adaptation for object detection also includes detection-specific methods that leverage the adversarial idea to align domains at multiple levels. For instance, Chen et al. (CVPR 2018) [13] introduce Domain Adaptive Faster R-CNN, which adds domain classifiers at both the image level and the instance level of Faster R-CNN to reduce domain discrepancies. Subsequently, Saito et al. (CVPR 2019) [14] propose SWDA (Strong-Weak Domain Alignment), which performs distribution alignment with two strategies: weak alignment at the global level (focusing on source–target images with similar layouts) and strong alignment at the local level (on small feature regions). Both methods train the detector within an adversarial-domain framework to extract domain-invariant features, improving performance on new datasets without requiring target-domain labels.

Building on this line of work, two recent representative methods serve as comparison baselines in this thesis. The first, **DA-Detect** [4], extends DA Faster R-CNN with an Adversarial Gradient Reversal Layer (AdvGRL) mechanism that performs hard-example mining together with an auxiliary domain generated by the RainMix random augmentation [42]. The second, **ALDI/ALDI++** [16], unifies feature alignment and Mean-Teacher self-distillation in a single framework, with a strategy that first stabilizes initialization and then performs distillation. Details of these two baselines and the implementation choices ensuring a fair comparison are presented in Section 4.2.3 of Chapter 4.

Chapter 3

Proposed method

FusionDA is built on top of an object detector and consists of two sequential phases, as illustrated in Figure 3.1 and Algorithms 1 and 2. *Phase 1 - Data Generation* narrows the domain gap at the input-image level: from a clear image $\mathbf{I}^s$, FusionDA synthesizes a "pseudo-foggy" image $\mathbf{I}^{sf}$; conversely, from a foggy image $\mathbf{I}^t$, FusionDA produces a "dehazed" image $\mathbf{I}^{tf}$. *Phase 2 - Training* uses all four image types to train FusionDA in a semi-supervised manner with coordinated components. The method employs the Mean Teacher framework and the Gradient Reversal Layer module, together with dedicated loss functions, to enhance the domain invariance of the learned features. All components are trained end-to-end through a single total loss function.

3.1 Data generation phase

The data generation phase builds two cross-domain translation streams based on a physical model and a pre-trained dehazing network, producing two cross-domain image pairs $(\mathbf{I}^s, \mathbf{I}^{sf})$ and $(\mathbf{I}^t, \mathbf{I}^{tf})$ that serve as input for the training phase in Section 3.2. Compared with adversarial image translators such as CycleGAN [43] and CUT [44], this approach requires no additional generative network and explicitly preserves scene structure through physical constraints. This is a key property because the synthesized images are directly reused with the ground-truth annotations of the original images during the training phase. An illustration is shown in Figure 3.2.

The key contribution of the data generation phase lies not merely in producing additional foggy data, but in the *same-scene, different-domain* property of the generated image pairs. Unlike arbitrary natural foggy images, the cross-domain images are produced from the original training images themselves and therefore fully share the scene structure and ground-truth annotations with the corresponding originals. This property allows FusionDA to directly reuse existing labels without any additional annotation, while ensuring that every difference within a pair stems purely from domain style without mixing content or semantics. This is precisely the condition required for the supervised losses and the feature-consistency constraint to operate as intended.

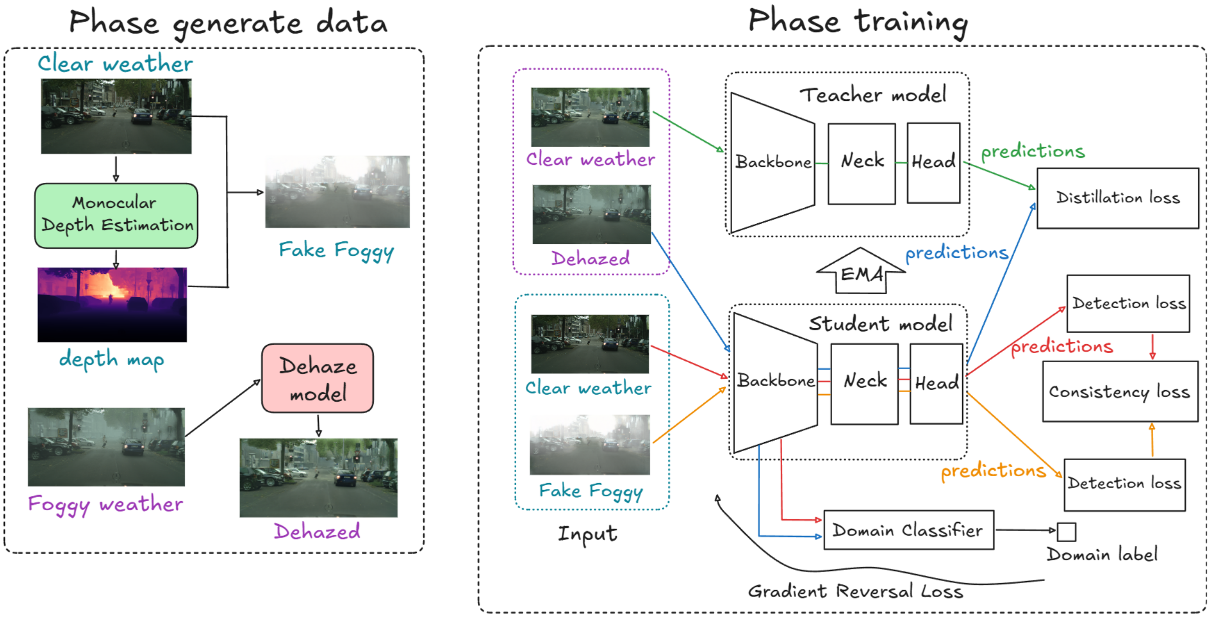

Figure 3.1: Overall architecture of FusionDA. **Data Generation Phase:** generates cross-domain images to narrow the domain gap at the image level. **Training Phase:** trains the model within the Mean Teacher framework combined with the GRL module.

Synthesizing the pseudo-foggy image $\mathbf{I}^{sf}$ via the atmospheric scattering model. The synthesis procedure consists of three sequential steps: (i) estimating a depth map from the clear image, (ii) computing a transmission map from the depth map, and (iii) applying the atmospheric scattering model to produce the pseudo-foggy image. Figure 3.3 illustrates the result of this procedure.

For each clear image $\mathbf{I}^s$, a pre-trained depth-estimation model is used to estimate the absolute depth map $d(x) \in \mathbb{R}^{H\times W}$ in meters, where x is the pixel coordinate and H, W are the image height and width. The transmission map $t(x)$ is then computed via Equation 3.1.

$$t(x) = e^{-\beta\cdot d(x)}. \tag{3.1}$$

where $t(x) \in [0, 1]$ is the fraction of light from the scene that reaches the sensor without being scattered at pixel x; $d(x)$ is the scene depth at pixel x in meters; and β is the atmospheric scattering coefficient.

Finally, FusionDA applies the atmospheric scattering model, and the pseudo-foggy image $\mathbf{I}^{sf}$ is synthesized according to Equation 3.2.

$$\mathbf{I}^{sf}(x) = \mathbf{I}^s(x)\cdot t(x) + A\big(1 - t(x)\big). \tag{3.2}$$

where $\mathbf{I}^s(x), \mathbf{I}^{sf}(x) \in \mathbb{R}^3$ are the RGB pixel intensities of the clear and pseudo-foggy images at coordinate x, respectively; $t(x)$ is given by Equation 3.1; and $A \in \mathbb{R}^3$ is the

Algorithm 1 Phase 1: Data Generation

Require: Clear domain dataset $\mathcal{D}_{\text{clear}} = \left\{(\mathbf{I}_i^s, \mathbf{B}_i^s, \mathbf{C}_i^s)\right\}$; Foggy domain dataset $\mathcal{D}_{\text{foggy}} = \left\{\mathbf{I}_j^t\right\}$; depth estimator $\mathcal{M}_d$; dehazing network G_{dh}; scattering coefficient β

Ensure: Paired-image queue $\mathcal{Q} = \mathcal{Q}_{\text{clear}} \cup \mathcal{Q}_{\text{foggy}}$

1: **for** each $\mathbf{I}^s \in \mathcal{D}_{\text{clear}}$ **do**
2: $\quad d \leftarrow \mathcal{M}_d(\mathbf{I}^s)$
3: $\quad \mathbf{I}^{sf} \leftarrow \text{AtmosphereScatterModel}(\mathbf{I}^s, d, \beta)$
4: **end for**
5: **for** each $\mathbf{I}^t \in \mathcal{D}_{\text{foggy}}$ **do**
6: $\quad \mathbf{I}^{tf} \leftarrow G_{\text{dehaze}}(\mathbf{I}^t)$
7: **end for**
8: Align by filename: $\mathcal{Q}_{clear} \leftarrow \{(\mathbf{I}^s, \mathbf{I}^{sf}, \mathbf{B}^s, \mathbf{C}^s)\}$, $\mathcal{Q}_{foggy} \leftarrow \{(\mathbf{I}^t, \mathbf{I}^{tf})\}$
9: **return** $\mathcal{Q} = \mathcal{Q}_{clear} \cup \mathcal{Q}_{foggy}$

atmospheric light intensity vector. Rather than fixing A, FusionDA estimates it from each clear image using the *dark channel prior* method of He et al. [45]. Specifically, the dark channel $J^{\text{dark}}(x)$ is computed according to Equation 3.3.

$$J^{\text{dark}}(x) = \min_{u \in \Omega(x)} \min_{c \in \{R,G,B\}} \mathbf{I}_c^s(u). \tag{3.3}$$

where $\Omega(x)$ is a 15×15-pixel neighborhood centered at x; u indexes pixels within $\Omega(x)$; c indexes the color channel; and $\mathbf{I}_c^s(u)$ is the value of channel c of the clear image at pixel u. Following the empirical recommendation of He et al. [45], the top 0.1% of pixels with the highest dark-channel values are selected as candidates for the atmospheric light, this ratio balances two opposing requirements: avoiding the selection of locally bright object regions (vehicle lights, reflective signs) while retaining enough points for a stable estimate of A. The final A is the RGB mean of the selected pixels in the original image.

Synthesizing the dehazed image $\mathbf{I}^{tf}$ via a dehazing network. In the reverse direction, FusionDA translates a foggy image $\mathbf{I}^t$ into the clear domain via a dehazing network. The output $\mathbf{I}^{tf}$ represents the content of the foggy image under clear-viewing conditions, close to the distribution of the clear domain, and is referred to as the dehazed image (see Figure 3.4).

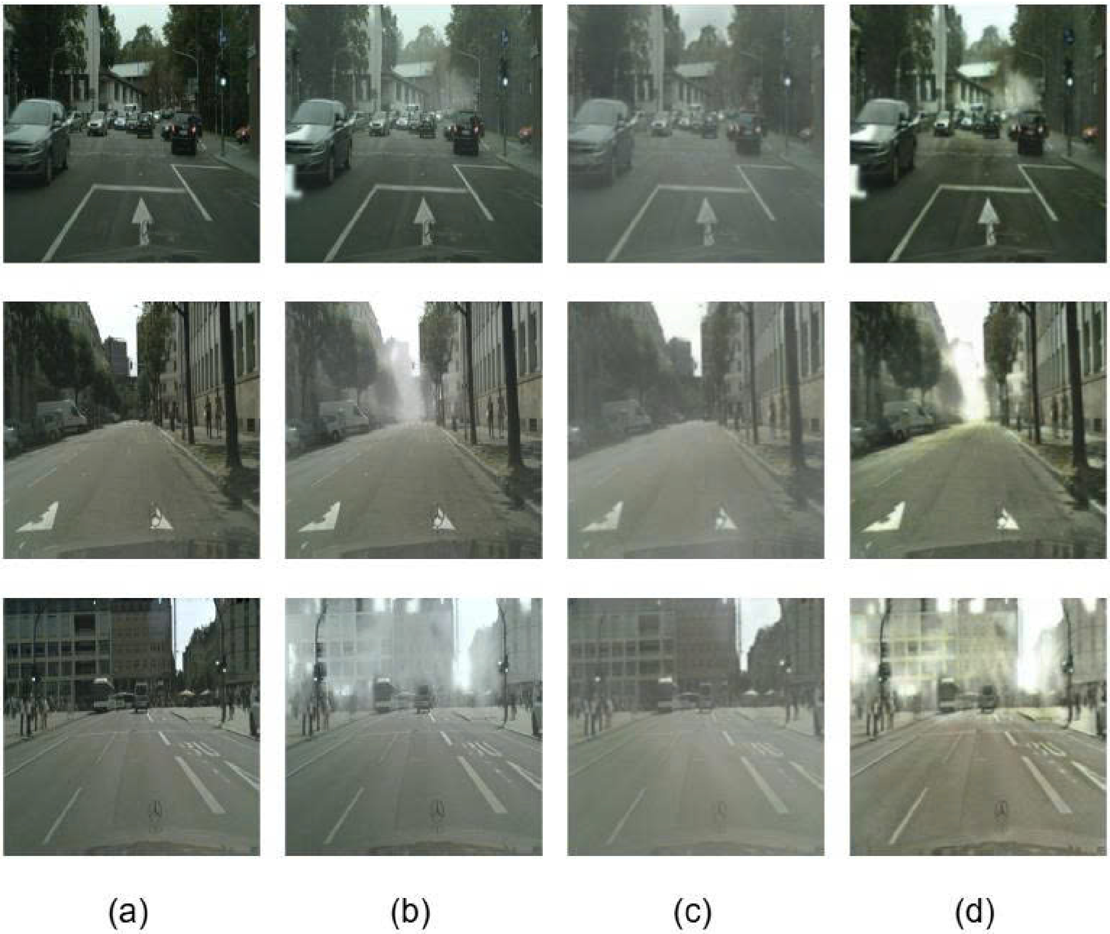

Figure 3.2: Illustration of the domain translation process, where (a) is the clear image $\mathbf{I}^s$, (b) is the foggy image $\mathbf{I}^t$, (c) is the pseudo-foggy image $\mathbf{I}^{sf}$, and (d) is the dehazed image $\mathbf{I}^{tf}$.

3.2 Training phase

The training phase pursues two complementary objectives: *exploiting* the unlabeled foggy images so that the model adapts to the foggy-domain distribution, while *preserving* detection performance on the clear domain, for which complete ground-truth annotations are already available. The core of FusionDA for these two objectives lies in combining two complementary mechanisms at two different levels: the *Mean Teacher framework* aligns at the output level through pseudo-labels from a stable teacher model, and the *Gradient Reversal Layer (GRL) module* aligns at the feature-space level through an adversarial mechanism. These two mechanisms are integrated end-to-end through a single total loss function with five components dedicated to each data source.

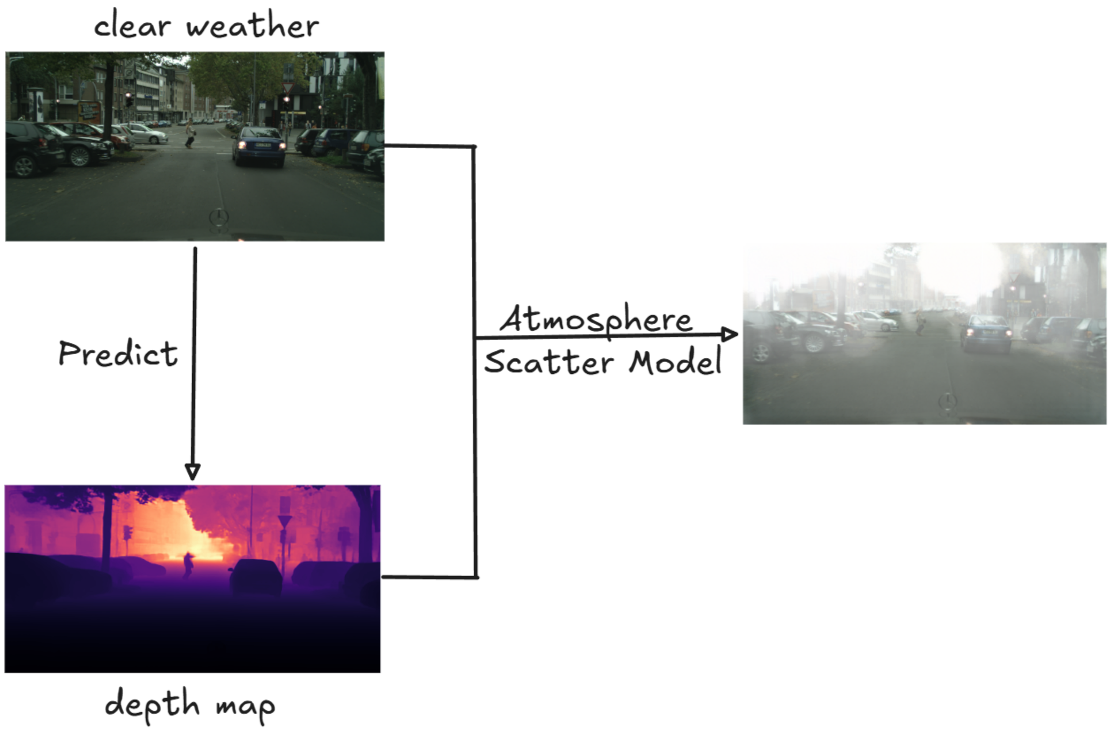

Figure 3.3: Illustration of the fog synthesis process applied to a clear image.

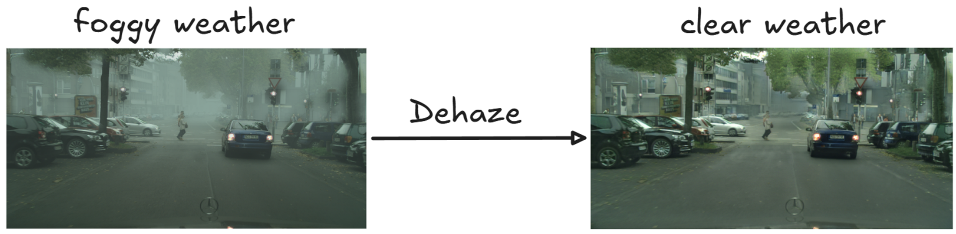

Figure 3.4: Illustration of the dehazing process.

3.2.1 Mean Teacher framework with the GRL module

Mean Teacher learning and distillation on unlabeled data. Mean Teacher (MT) [24] is a semi-supervised learning framework consisting of two models with the same architecture but different update rules: the *student* is optimized via ordinary gradient descent, while the *teacher* is updated as an exponential moving average (EMA) of the student's weights. This design provides two key benefits for DAOD: (i) the slowly updated teacher produces highly stable predictions that are reliable enough to serve as pseudo-labels on unlabeled data; and (ii) the student learns from a supervisory signal derived from its own stable counterpart, which reduces prediction variance and improves robustness. Concretely, letting $\mathbf{P}_{\text{student}}$ and $\mathbf{P}_{\text{teacher}}$ denote the weight parameters of the student and teacher, respectively, the teacher's weights are updated at every training batch according to Equation 3.4.

$$\mathbf{P}_{\text{teacher}} = \alpha\, \mathbf{P}_{\text{teacher}} + (1-\alpha)\, \mathbf{P}_{\text{student}}. \tag{3.4}$$

where $\alpha \in [0, 1)$ is the decay coefficient.

When MT is integrated into FusionDA, the teacher receives the dehazed images $\mathbf{I}^{tf}$, produced from the original foggy images by the dehazing network in the Data Generation Phase. This lets the teacher predict under clear-viewing conditions that are closer to the distribution it has seen during training. High-confidence bounding boxes from these predictions are filtered into pseudo-labels, which are then used to supervise the student on the corresponding original foggy images $\bar{\mathbf{I}}^{t}$. This "teacher-sees-clean, student-sees-foggy" mechanism allows FusionDA to transfer cross-domain knowledge without requiring ground-truth annotations on the foggy images.

GRL learns domain-invariant features through adversarial training. The GRL module [22] complements the MT framework with a more general mechanism: rather than aligning only at the output level (pseudo-labels), the GRL directly forces the backbone's *feature space* to be invariant across the two domains. This learning scheme is illustrated in Figure 3.5. In the proposed method, the feature extractor $\mathcal{G}_f$ is the model's backbone, while the label predictor $\mathcal{G}_y$ comprises the neck and head that produce the final predictions. The core idea is to attach a domain classifier G_d after the backbone and train it adversarially (by multiplying the propagated gradient by a factor of $-\lambda$) so that G_d cannot distinguish features originating from clear or foggy images. The backbone is thereby forced to learn truly domain-invariant representations, a form of implicit supervision that MT cannot provide, since MT operates only on the final detection outputs.

In FusionDA, the domain classifier G_d is designed in the modern DANN style with a *GAP + MLP* architecture [13] to ensure stable gradients during adversarial training. The detailed architecture is described in the experiments. The domain loss uses Binary Cross-Entropy (BCE) combined with *label smoothing* at $\varepsilon = 0.1$. Instead of hard labels $\{0, 1\}$, soft labels are defined as in Equations 3.5 and 3.6.

$$y_s = 1 - \varepsilon = 0.9 \quad \text{(clear domain)} \tag{3.5}$$

$$y_t = \varepsilon = 0.1 \quad \text{(foggy domain)} \tag{3.6}$$

Label smoothing [46] prevents the domain classifier from becoming overconfident, flattens

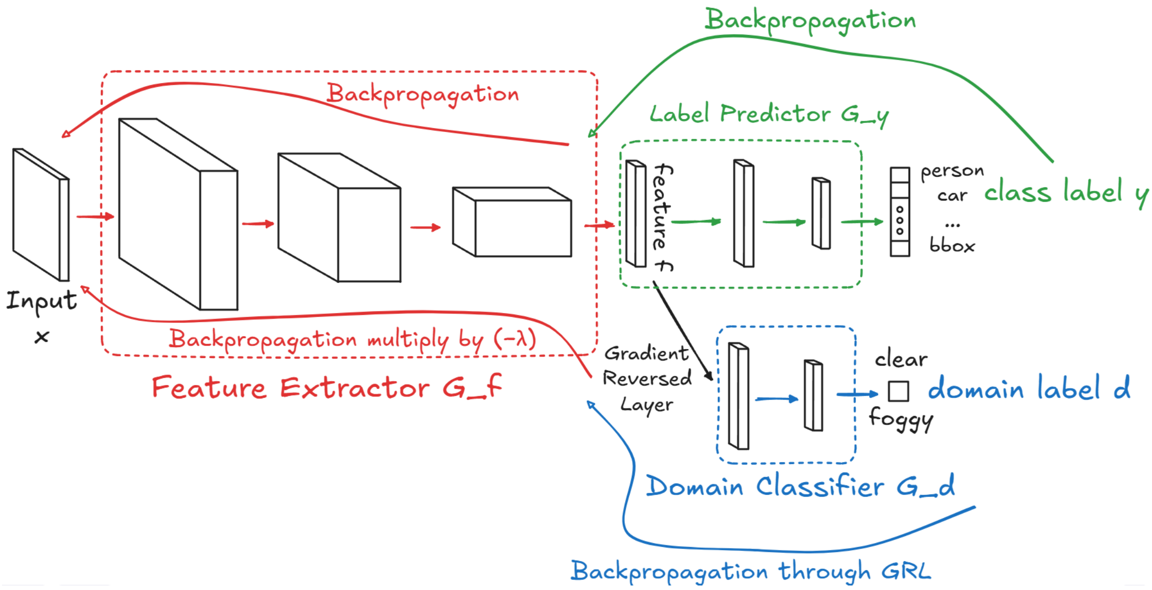

Figure 3.5: Illustration of domain-invariant feature learning via the Gradient Reversal Layer (GRL). The model has three parts: the feature extractor $\mathcal{G}_f$, the label predictor $\mathcal{G}_y$, and the domain classifier $\mathcal{G}_d$. The GRL reverses the gradient from the domain-discrimination branch, forcing $\mathcal{G}_f$ to learn features that cannot be distinguished between the clear and foggy domains. This lets the model learn shared features, reducing the impact of domain differences and improving generalization.

the output probability distribution, and yields richer gradient signals throughout adversarial training. This is particularly important for foggy-domain adaptation, where the boundary between the two domains is already blurred.

Complementarity between MT and GRL. The two mechanisms complement each other at two levels of domain alignment. MT aligns at the *output level* via pseudo-labels, which is useful when accurate detection signals can be recovered as long as the input image is close enough to the training distribution. GRL aligns at the *feature level* and is particularly effective when the backbone must generalize to both domains simultaneously, even in cases where MT's pseudo-labels may be noisy or incomplete. Combining both within a single model allows FusionDA to fully exploit the potential of the unlabeled foggy data at both levels. The quantitative contribution of each mechanism is validated by the *ablation study* in Chapter 4.

3.2.2 Total loss function

FusionDA is trained *end-to-end* by jointly optimizing all the supervisory signals introduced above through a single total loss function, defined in Equation 3.7.

$$\mathbf{L} = \underbrace{\mathbf{L}_{det}(\mathbf{I}^s)}_{\text{clear}} + \underbrace{w_{sf}(p) \cdot \mathbf{L}_{det}(\mathbf{I}^{sf})}_{\text{pseudo-foggy}} + \underbrace{w_{\text{dis}} \cdot \mathbf{L}_{dis}(\mathbf{I}^{tf}, \mathbf{I}^t)}_{\text{distillation}} + \underbrace{w_{\text{con}} \cdot \mathbf{L}_{con}}_{\text{feature consistency}} + \underbrace{w_{\text{dom}} \cdot \mathbf{L}_{dom}}_{\text{GRL}}. \tag{3.7}$$

where $w_{sf}(p), w_{dis}, w_{con}, w_{dom}$ are balancing weights determined empirically and $p \in [0, 1]$ denotes the training progress.

$\mathbf{L}_{det}(\mathbf{I}^s)$ - Detection loss on the clear domain. This is the foundational supervisory signal of FusionDA, ensuring that the model retains detection capability on images with full ground-truth annotations, and is computed via Equation 3.8.

$$\mathbf{L}_{det}(\mathbf{I}^s) = \mathbf{L}_{box}(\mathbf{I}^s, \mathbf{B}^s) + \mathbf{L}_{cls}(\mathbf{I}^s, \mathbf{C}^s) + \mathbf{L}_{dfl}(\mathbf{I}^s, \mathbf{C}^s). \tag{3.8}$$

where $\mathbf{B}^s$ and $\mathbf{C}^s$ are the ground-truth bounding boxes and corresponding object classes for image $\mathbf{I}^s$, respectively. $\mathbf{L}_{box}$, $\mathbf{L}_{cls}$, and $\mathbf{L}_{dfl}$ are the bounding-box regression loss, the classification loss, and the distribution focal loss, respectively.

$\mathbf{L}_{det}(\mathbf{I}^{sf})$ - Detection loss on pseudo-foggy images. To counter the student's bias toward clear images, FusionDA adds a new supervised training branch whose input consists of the pseudo-foggy images $\mathbf{I}^{sf}$ generated in the Data Generation Phase. Because $\mathbf{I}^{sf}$ shares the scene structure and therefore the ground-truth labels with $\mathbf{I}^s$, the annotations $\mathbf{B}^{sf}, \mathbf{C}^{sf}$ are directly reused from $\mathbf{B}^s, \mathbf{C}^s$. This is an advantage that GAN-based methods (CycleGAN, CUT) cannot offer because they do not preserve scene structure. The loss is computed according to Equation 3.9.

$$\mathbf{L}_{det}(\mathbf{I}^{sf}) = \mathbf{L}_{box}(\mathbf{I}^{sf}, \mathbf{B}^{sf}) + \mathbf{L}_{cls}(\mathbf{I}^{sf}, \mathbf{C}^{sf}) + \mathbf{L}_{dfl}(\mathbf{I}^{sf}, \mathbf{C}^{sf}). \tag{3.9}$$

The weight $w_{sf}(p)$ of this loss decays linearly with training progress, gradually reducing the influence of style noise in the pseudo-foggy images during the final stage, when the model has stabilized on the real foggy domain.

$\mathbf{L}_{dis}$ - Knowledge distillation from the teacher on unlabeled data. Following the Mean Teacher mechanism described in Section 3.2.1, the teacher predicts on the dehazed image $\mathbf{I}^{tf}$ to generate pseudo-labels, which are then used to supervise the student on the corresponding foggy image $\bar{\mathbf{I}}^{t}$. The distillation loss applied during training is defined in Equation 3.10.

$$\mathbf{L}_{dis}(\mathbf{I}^{tf}, \mathbf{I}^{t}) = \mathbf{L}_{det}\Big(\mathbf{I}^{t},\ \mathbf{F}_{\mathbf{B}}[\mathbf{T}_{\mathbf{B}}(\mathbf{I}^{tf})],\ \mathbf{F}_{\mathbf{C}}[\mathbf{T}_{\mathbf{C}}(\mathbf{I}^{tf})]\Big)\,. \tag{3.10}$$

where $\mathbf{T}_B(\cdot)$ and $\mathbf{T}_C(\cdot)$ are the teacher's raw bounding-box and label predictions on image $\mathbf{I}^{tf}$. These raw predictions then pass through the filters $\mathbf{F}_B[\cdot]$ and $\mathbf{F}_C[\cdot]$ to retain only the most reliable pseudo-labels above a confidence threshold, with the threshold determined empirically.

$\mathbf{L}_{con}$ - Feature-consistency constraint at the backbone level. While $\mathbf{L}_{dis}$ aligns at the output level, FusionDA further imposes a constraint *at the feature level* through the consistency loss in Equation 3.11. Exploiting the fact that the image pair $(\mathbf{I}^{s}, \mathbf{I}^{sf})$ shares the same scene structure but differs in domain style, FusionDA forces the backbone to produce similar feature representations on both input streams.

$$\mathbf{L}_{con} = 1 - \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \cos\Big(\phi_{\text{student}}(\mathbf{I}_i^{sf}),\ \phi_{\text{student}}(\mathbf{I}_i^{s})\Big)\,. \tag{3.11}$$

where $\mathcal{B}$ is the set of sample indices in the current mini-batch; $\phi_{\text{student}}(\cdot)$ is the output feature vector of the student backbone; and $\cos(\cdot,\cdot)$ is the cosine similarity between two feature vectors. The gradient from this loss propagates only through the $\mathbf{I}^{sf}$ branch to avoid redundancy, since the $\mathbf{I}^{s}$ branch is already updated by the loss in Equation 3.8.

$\mathbf{L}_{dom}$ - Global adversarial loss from the GRL module. This loss operates at the *global level*, forcing the backbone to learn features that the domain classifier G_d cannot distinguish, as in Equation 3.12. The loss is averaged over the two domains using the soft labels introduced in Section 3.2.1.

$$\mathbf{L}_{dom} = \frac{1}{2}\Big[\mathbf{L}_{\text{BCE}}(G_d(\phi_{\text{student}}(\mathbf{I}^{s})),\ y_s) + \mathbf{L}_{\text{BCE}}\big(G_d(\phi_{\text{student}}(\mathbf{I}^{sf})),\ y_t\big)\Big]\,. \tag{3.12}$$

where $G_d(\cdot)$ is the domain classifier with a simple MLP architecture preceded by a global feature-extraction layer. y_s and y_t are the soft domain labels corresponding to the input images (Equations 3.5 and 3.6). When the optimization converges and G_d can no longer distinguish between the two domains (domain accuracy approaches 0.5), the backbone is confirmed to have achieved genuine domain invariance.

Algorithm 2 Phase 2: Training

Require: Queue $\mathcal{Q}$ from Algorithm 1; student θ_{Student}; domain classifier G_d; threshold τ; loss weights $w_{sf}(p)$, w_{dis}, w_{con}, w_{dom}; total epochs E

Ensure: Domain-adapted student parameters θ_{Student}

Initialise teacher: $\theta_{\text{Teacher}} \leftarrow \theta_{\text{Student}}$ ▷ frozen-gradient copy
for $e = 1$ **to** E **do**
  for each mini-batch $(\mathbf{I}^s, \mathbf{I}^{sf}, \mathbf{I}^t, \mathbf{I}^{tf}, \mathbf{B}^s, \mathbf{C}^s) \sim \mathcal{Q}$ **do**
    $(\hat{y}_A,\ \phi_A) \leftarrow \theta_{\text{Student}}(\mathbf{I}^s)$ ▷ Stream A: student on clear image
    $(\hat{y}_B,\ \phi_B) \leftarrow \theta_{\text{Student}}(\mathbf{I}^{sf})$ ▷ Stream B: student on pseudo-foggy image
    $(\hat{y}_C,\ \phi_C) \leftarrow \theta_{\text{Student}}(\mathbf{I}^t)$ ▷ Stream C: student on foggy image
    $\hat{y}_D \leftarrow \theta_{\text{Teacher}}(\mathbf{I}^{tf})$ **[no grad]** ▷ Stream D: teacher on dehazed image
    Apply NMS to $\hat{y}_D$ with confidence threshold τ and filtering $\rightarrow \tilde{\mathbf{B}}^t$, $\tilde{\mathbf{C}}^t$
    $\mathbf{L}_{\text{s}} \leftarrow \mathbf{L}_{\text{det}}(\hat{y}_A,\ \mathbf{B}^s, \mathbf{C}^s)$ ▷ Eq. 3.8
    $\mathbf{L}_{sf} \leftarrow \mathbf{L}_{\text{det}}(\hat{y}_B,\ \mathbf{B}^s, \mathbf{C}^s)$ ▷ Eq. 3.9
    if past burn-in epoch **then**
      $\mathbf{L}_{\text{dis}} \leftarrow \mathbf{L}_{\text{det}}(\hat{y}_C,\ \tilde{\mathbf{B}}^t, \tilde{\mathbf{C}}^t)$ ▷ Eq. 3.10
      $\mathbf{L}_{\text{con}} \leftarrow 1 - \cos\big(\phi_B,\ \phi_A\big)$ ▷ Eq. 3.11
    else
      $\mathbf{L}_{\text{dis}}$, $\mathbf{L}_{\text{con}} \leftarrow 0$
    end if
    if past GRL warm-up epoch **then**
      $\mathbf{L}_{\text{BCE}} \leftarrow \mathbf{L}_{\text{adv}}(G_d(\phi_A),\ y_s;\ G_d(\phi_B),\ y_t)$ ▷ Eq. 3.12
    else
      $\mathbf{L}_{\text{dom}} \leftarrow 0$
    end if
    $\mathbf{L} \leftarrow \mathbf{L}_{\text{src}} + w_{sf}(p)\cdot\mathbf{L}_{sf} + w_{\text{dis}}\cdot\mathbf{L}_{\text{dis}} + w_{\text{con}}\cdot\mathbf{L}_{\text{con}} + w_{\text{dom}}\cdot\mathbf{L}_{\text{dom}}$ ▷ Eq. 3.7
    Back-propagate and update θ_{Student} and G_d; reset gradients
    if past EMA delay **then**
      $\theta_{\text{Teacher}} \leftarrow \alpha\,\theta_{\text{Teacher}} + (1-\alpha)\theta_{\text{Student}}$ ▷ Eq. 3.4
    end if
  end for
end for
return θ_S

Chapter 4

Experiments

4.1 Research questions

To systematically evaluate the effectiveness of the proposed FusionDA method, we design experiments around three main research questions:

- **RQ1 - Overall effectiveness of FusionDA:** Does the proposed FusionDA method generalize better than representative domain-adaptive object detection (DAOD) methods on standard benchmarks?
- **RQ2 - Component analysis of FusionDA:** How does each component integrated into FusionDA contribute to the final performance?
- **RQ3 - Generalization of FusionDA:** Does the method remain effective when applied to a detection task with a different label structure and a different type of fog corruption?

RQ1 aims to position FusionDA among modern DAOD methods; RQ2 dissects and analyzes the role of each proposed component, thereby verifying the design hypotheses presented in Chapter 3; while RQ3 serves as a sanity check to confirm that the proposed mechanisms are independent of the Cityscapes/Foggy Cityscapes benchmark.

4.2 Experimental setup

4.2.1 Datasets

The method is evaluated on standard datasets spanning three domains: clear, synthetic fog, and natural fog. The datasets are deliberately chosen to complement each other to maximize the generality of the results.

Cityscapes [25] consists of 2,975 training images and 500 validation images, capturing urban street scenes under normal weather, collected from 50 cities.

Table 4.1: Summary of datasets used in the experiments. All are standard datasets widely used and evaluated in object detection and domain adaptation.

Dataset	# Train images	# Val images	Description
Cityscapes [25]	2,975	500	Clear weather, urban street scenes
Foggy Cityscapes [9] ($\beta = 0.02$)	2,975	500	Synthetic fog (derived from Cityscapes)
RTTS [27]	0	4,322	Natural fog, diverse outdoor scenes
Foggy Driving [9]	0	101	Natural fog, driving scenes
FoggyZurich-test [28]	0	40	Dense natural fog, driving scenes
WIDERFACE-easy [29]	12,880	3,226	Face detection (supplementary RQ3 experiment)

Foggy Cityscapes [9] is created by adding synthetic fog to the Cityscapes images. Foggy Cityscapes therefore shares the same train/validation split as Cityscapes. During training, we use the Cityscapes training set together with the unlabeled training set of Foggy Cityscapes, while evaluation is performed on the Foggy Cityscapes validation set. The experiments adopt the dense fog level corresponding to $\beta = 0.02$.

RTTS [27] is a relatively comprehensive dataset collected under natural foggy conditions, containing 4,322 real-fog images annotated for five object classes. Because collecting paired clear and foggy images in the real world is extremely difficult, nearly impossible, Li et al. proposed RTTS to evaluate the real-world generalization of models.

Foggy Driving [9] contains 101 color images depicting driving scenes under natural fog. Among them, 51 images were captured with a mobile-phone camera in foggy conditions across various areas of Zürich, and the remaining 50 were collected from the internet.

FoggyZurich [28] is a dataset of driving images under natural fog containing 3,808 street scenes collected in Zurich and surrounding areas with an on-vehicle camera under naturally foggy conditions. In this study, the FoggyZurich-test subset of 40 images with high fog density is used to evaluate object detection in real-world foggy environments.

WIDERFACE [29] is a face-detection benchmark consisting of 16,106 images with diverse face cases. The experiments use the EASY subset, which contains faces that are easier to detect (larger average size, lower occlusion). This set serves as the clear domain in the supplementary RQ3 experiment, while the foggy domain is generated directly using the `imagecorruptions` library [30] (Bethge Lab). An illustration is shown in Figure 4.1. This corruption-generation procedure differs fundamentally from the FusionDA setup on Foggy Cityscapes (which is based on the atmospheric scattering model), creating an opportunity to verify the independence of FusionDA from the type of fog corruption.

Figure 4.1: Sample images from the WIDERFACE dataset.

In the main experiments, the model is evaluated on only two object classes: *person* and *car*. These two classes occur with high and stable frequency across Cityscapes, Foggy Cityscapes, and the natural-fog datasets, ensuring a sufficient sample size for reliable evaluation. In addition, *person* and *car* are critical objects in intelligent-transportation and driver-assistance tasks, particularly under adverse weather such as fog.

The chosen datasets allow us to evaluate FusionDA's generalization to the foggy domain using only unlabeled foggy data. This faithfully reflects the practical scenario of deploying the model in a new environment where no labeled data is available.

4.2.2 Implementation

All experiments are implemented in Python and PyTorch [47] and the Ultralytics YOLO library. Training and evaluation are conducted on a server equipped with an NVIDIA RTX 4080S GPU (16 GB VRAM), CUDA 12.4 and cuDNN 12.4. The operating system is Ubuntu 22.04 LTS.

The source code of this thesis is organized in a modular structure to facilitate reproduction and extension. The main modules are built on top of established repositories from the research community and on the documentation released by the original authors. During development, the source code was refactored to improve readability and maintainability and to ensure consistency across components. In addition, all key modules are accompanied by detailed comments to facilitate tracing and reuse in subsequent research.

To ensure reproducibility, the full source code, model checkpoints, configuration files, and datasets are publicly released. The source code is hosted on GitHub 1. Setup instructions for the environment, training, and evaluation are provided in detail in the repository's

README.md. The trained checkpoints and the datasets used in the experiments are stored on Google Drive 2.

4.2.3 Comparison methods

For an objective and comprehensive evaluation of FusionDA, we compare against the standard fine-tuning baseline described in Chapter 2 and two representative methods in domain-adaptive object detection (DAOD): DA-Detect [4] and ALDI [16]. These two methods represent the two dominant DAOD paradigms: adversarial feature alignment and Mean-Teacher self-distillation. Moreover, both are validated on the standard Cityscapes/-Foggy Cityscapes pair. In addition, both release publicly available source code, enabling faithful result reproduction under identical experimental conditions.

Standard fine-tuning (Full-finetune)

In fine-tuning, models are trained in a supervised manner on labeled clear images from Cityscapes. The models are initialized with weights pre-trained on COCO. Fine-tuning therefore leverages the generic features the model has previously learned to accelerate convergence and produce an initial baseline configuration. These baseline configurations reflect the drop in detection performance when the model is transferred from the clear domain to the foggy domain. The performance gap between the fine-tuned configurations and the corresponding DAOD methods directly reflects the effectiveness of domain-adaptation strategies in improving generalization to the foggy domain. Beyond serving as the lower-bound baseline, these configurations also enable a direct assessment of how much each domain-adaptation component in FusionDA contributes, and are crucial for analyzing the domain gap between clear and foggy images.

DA-Detect

DA-Detect [4] is an unsupervised domain-adaptation method built on the Faster R-CNN architecture, targeting object detection under foggy conditions for autonomous driving.

In our experiments, DA-Detect is implemented on the common Faster R-CNN backbone (ResNet-50 [48] with Feature Pyramid Network, R50-FPN). This is also the shared backbone used to ensure fairness when comparing against the proposed method. Two variants are evaluated: (i) the model trained only on labeled clear images (base), which serves as the lower bound when no domain adaptation is applied; and (ii) the full DA-Detect with all domain-adaptation components enabled. Compared to FusionDA, DA-Detect differs in that it generates an auxiliary domain via random augmentation rather than structurally consistent cross-domain image pairs based on a physical model.

ALDI

ALDI [16] is a unified DAOD framework that self-distillation within a single student-teacher architecture with exponential moving average (EMA) updates.

While both FusionDA and ALDI share the core idea of a Mean Teacher framework with EMA updates, FusionDA differs from ALDI in two key respects: (a) it adds a cross-domain image synthesis pipeline based on the atmospheric scattering model and the AODNet dehazing network, rather than relying solely on standard data augmentations to narrow the domain gap at the image level; and (b) it integrates an additional adversarial Gradient Reversal Layer module that explicitly forces the backbone to learn domain-invariant features.

ALDI++

ALDI++ is an improved version of ALDI, focusing on enhancing the quality of self-training and the stability of training. Compared with the original ALDI, ALDI++ adds strong augmentations, uses an EMA model to initialize a more stable teacher, and applies a soft-distillation mechanism that allows the student to learn directly from the teacher's probability distribution without a fixed confidence threshold. The method also reduces reliance on feature alignment to improve training stability while maintaining high performance.

To ensure a fair comparison with FusionDA, which is built on the one-stage YOLO26 architecture, ALDI++ is implemented on two configurations: R50-FPN within the Faster R-CNN framework, and YOLOv5m. As with DA-Detect, each configuration is evaluated in two variants: clear-only training and the full ALDI++.

4.2.4 Architectures and hyperparameters

Data generation models

The data generation phase of FusionDA requires two specialized models: a depth-estimation model to synthesize the pseudo-foggy image $\mathbf{I}^{sf}$ through the atmospheric scattering model, and a dehazing network to generate the dehazed image $\mathbf{I}^{tf}$ from a foggy image. The choice of these models must balance three criteria: *output quality* sufficient to narrow the domain gap at the input-image level, *inference cost* low enough to process the entire training set, and *reproducibility* through publicly available checkpoints that require no further fine-tuning. It is important to emphasize that this is a deliberate choice: $\mathbf{I}^{sf}$ and $\mathbf{I}^{tf}$ do not serve as final outputs but as intermediate inputs to the training phase. Photorealistic quality is therefore not required; a *good enough* generator must preserve scene structure and push the pixel distribution toward the target domain.

Depth-estimation model: DepthAnythingV2. The absolute depth map $d(x) \in \mathbb{R}^{H\times W}$ in meters is estimated using DepthAnythingV2 [49] fine-tuned on Virtual KITTI 2 [50]. The synthesis pipeline is in principle compatible with any depth-estimation model capable of producing absolute depth. Among the candidates evaluated for stability and optimality, FusionDA chooses DepthAnythingV2 for several reasons. First, SOTA performance: the model achieves $\delta_1 = 95.0\%$ on KITTI, outperforming ZoeDepth, MiDaS, and Marigold [49]. Second, distribution compatibility: DepthAnythingV2 provides a variant fine-tuned on Virtual KITTI 2 that covers urban scenes similar to Cityscapes, ensuring reliable absolute-depth estimation. Finally, the model is open-source, has fast inference, and requires no additional fine-tuning. To increase the diversity of the synthesized data, the scattering coefficient β is sampled uniformly as $\beta \sim \mathcal{U}(0.005, 0.015)$ per image (Foggy Cityscapes uses $\beta = 0.02$), corresponding to light-to-medium fog over scenes with depth up to 80 m.

Dehazing network: AODNet. In the reverse direction, FusionDA translates a foggy image $\mathbf{I}^t$ to the clear domain via AODNet [5]. AODNet directly estimates the dehazed image $\mathbf{I}^{tf}$ from the input through a lightweight convolutional network, bypassing the intermediate steps of estimating a transmission map or atmospheric light. As with depth estimation, the procedure is in principle compatible with any dehazing network that produces a dehazed image. We adopt this model because it is lightweight (1.7K parameters), providing near-instantaneous inference on the entire Foggy Cityscapes set without requiring specialized hardware acceleration. AODNet also avoids error propagation by estimating the dehazed image directly, eliminating the risk of error accumulation through multiple intermediate steps. The model further provides a checkpoint pre-trained on the outdoor-fog RESIDE dataset, which is widely validated and publicly available. More recent SOTA models such as FFA-Net [6], DehazeFormer [7], and C2PNet [8] achieve substantially higher PSNR, but at much higher computational cost; their perceptual-quality advantage is not needed here because $\mathbf{I}^{tf}$ only serves as the input to the teacher model in the Mean Teacher framework (Section 3.2.1), not as the final image presented to the user.

GRL module and domain classifier architecture

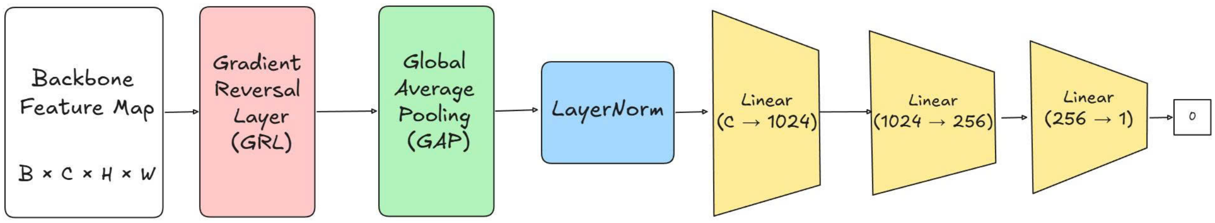

Figure 4.2: Architecture of the domain classifier module in FusionDA: GAP reduces the spatial feature map to a global representation, and an MLP with LayerNorm performs domain classification.

The domain classifier G_d in FusionDA follows the *GAP + MLP* architecture: the backbone feature map $\phi \in \mathbb{R}^{B\times C\times H\times W}$ is reduced to a global representation $[B, C]$ via Global Average Pooling, and then classified by a three-layer MLP with LayerNorm and a binary-logit output. This image-level discriminator design following the GAP + FC convention was established by Chen et al. [13] as one of the foundations of modern DAOD, and is inspired by Adaptive Teacher [15]. The architecture of G_d in FusionDA is therefore not an arbitrary choice but reflects the consensus of the DAOD community.

Hyperparameter values

Table 4.2 summarizes the key hyperparameters of FusionDA, determined empirically. The following analysis discusses the rationale behind each group of parameters.

Table 4.2: Key hyperparameters of FusionDA.

Type	**Hyperparameter**	**Symbol**	**Value**
Training	Initial learning rate	η_0	10^{-4}
	Final learning rate	η_f	10^{-2} $(\times\eta_0)$
	Weight decay	—	5×10^{-4}
	Gradient clipping	—	2.0
Mean Teacher	EMA coefficient	α	0.9999
	Pseudo-label threshold	τ	0.50
Loss	Weight $\mathbf{L}_{det}(\mathbf{I}^{sf})$ (init)	w^0_{sf}	0.10
	Weight $\mathbf{L}_{dis}$	w_{dis}	0.20
	Weight $\mathbf{L}_{con}$	w_{con}	0.50
	Weight $\mathbf{L}_{dom}$	w_{dom}	0.05
GRL	Max adversarial coefficient	λ_{max}	1.0
	Label smoothing	ε	0.1

Learning rate and optimizer. FusionDA uses AdamW [51] with $\eta_0 = 10^{-4}$ and weight decay 5×10^{-4}. The learning rate decays following a cosine schedule to $\eta_f = 0.01 \times \eta_0$ at the final epoch. Gradient clipping at the threshold 2.0 is applied to both the student model and the discriminator G_d after each backward pass, suppressing local gradient spikes that often occur in the early stage when GRL is first activated and destabilizes the backbone.

EMA coefficient and pseudo-label threshold. The EMA coefficient $\alpha = 0.9999$ is adopted from Adaptive Teacher [15] and ALDI++ [16]: this value keeps the teacher updating very slowly, reflecting only the long-term trend of the student rather than per-batch fluctuations, thereby producing more stable pseudo-labels. The pseudo-label threshold $\tau = 0.50$ filters out low-confidence boxes, high enough to retain label quality and yet low enough to provide a sufficient number of labels for learning.

Loss weights. The weight $w_{sf}(p)$ of $\mathbf{L}_{det}(\mathbf{I}^{sf})$ decays linearly with the training progress $p \in [0, 1]$ from $w^0_{sf} = 0.10$ down to 0.02 at the final epoch, following Equation 4.1.

$$w_{sf}(p) = w^0_{sf} \cdot \max\big(0.2,\ 1 - 0.8p\big). \tag{4.1}$$

In the early stage, pseudo-foggy images provide useful augmentation. Toward the end, when the model has adapted well to the foggy domain via $\mathbf{L}_{dis}$, the influence of $\mathbf{I}^{sf}$ is gradually reduced to avoid style noise from the scattering model. The consistency weight $w_{con} = 0.50$ is larger than $w_{dis} = 0.20$ because $\mathbf{L}_{con}$ operates only on image pairs sharing the same ground-truth labels $(\mathbf{I}^s, \mathbf{I}^{sf})$, so its gradient signal is more reliable and must be strong enough to force the backbone toward domain invariance at the feature level. The weight $w_{dom} = 0.05$ keeps $\mathbf{L}_{dom}$ at a level sufficient for the GRL to operate without overwhelming $\mathbf{L}_{det}(\mathbf{I}^s)$. In preliminary experiments, $w_{dom} \geq 0.2$ caused the student model to lose detection capability on the clear domain. These weights were thus tuned and selected through experiments based on the observed magnitudes of the loss functions. This calibration ensures that the method aligns properly with its objectives.

Adversarial coefficient and label smoothing. The adversarial coefficient λ increases following the sigmoid schedule of Ganin et al. [22] from 0 after the warm-up phase, ensuring that the backbone has converged to a certain level before the adversarial signal is introduced. Label smoothing with $\varepsilon = 0.1$ is applied to the domain loss (Equations 3.5–3.6). In experiments using hard labels $\{0, 1\}$, the discriminator converges too quickly and the gradient flowing back through the GRL saturates. Setting $\varepsilon = 0.1$ keeps the output probabilities within the range $(0.1, 0.9)$, maintaining rich gradients throughout adversarial training.

4.2.5 Summary of experimental configurations

In total, the study compares FusionDA against the baseline configurations listed in Table 4.3.

To ensure data fairness, all configurations are trained on the labeled training set of Cityscapes and the unlabeled training set of Foggy Cityscapes, except for the standard fine-tuning configurations, which use only the Cityscapes training set. The configurations are then evaluated on the validation sets of Cityscapes, Foggy Cityscapes, and the natural-fog datasets.

The proposed solution is compared with other methods using the same pre-trained backbone for further training. Specifically, the proposed method uses the R50-FPN backbone to compare against fine-tuning, DA-Detect, and ALDI++; uses YOLOv5m to compare against ALDI++; and uses the YOLO26 family to compare against fine-tuning. This ensures fairness with respect to the initial model when comparing among DAOD methods.

Table 4.3 summarizes the main training hyperparameters of each configuration, including the training set, backbone, input image size, batch size, and the corresponding number of epochs. All methods are trained on the same hardware to ensure objectivity when comparing training times.

Table 4.3: Summary of training hyperparameters of the methods compared in RQ1. (L) and (U) denote labeled and unlabeled sets, respectively; the *Augment* column marks configurations that use augmented data (FusionDA's cross-domain images $\mathbf{I}^{sf}$, $\mathbf{I}^{tf}$ or DA-Detect's RainMix).

Method	**Training set**			**Backbone**	**Input**	**Batch**	**Epoch / Iter**
	Cityscapes (L)	**Foggy Cityscapes (U)**	**Augment**				
DA-Detect base	✓			R50-FPN	640	4	28000 iters
DA-Detect [4]	✓	✓	✓	R50-FPN	640	4	28000 iters
ALDI	✓			R50-FPN	640	16	14000 iters
ALDI++ [16]	✓	✓		R50-FPN	640	16	14000 iters
ALDI	✓			YOLOv5-m	640	16	14000 iters
ALDI++	✓	✓		YOLOv5-m	640	16	14000 iters
Full-finetune	✓			R50-FPN, YOLOv5-m, YOLO26-{n, s, m, l, x}	640	4	40 epochs
Ours (FusionDA)	✓	✓	✓	R50-FPN, YOLOv5-m, YOLO26-{n, s, m, l, x}	640	4	40 epochs

4.3 Evaluation metrics

We adopt mean average precision (mAP) as the main and most widely used metric. When evaluating object detection models, we are interested not only in classification but also in correctly localizing the bounding box overlapping with the true object. Simple metrics such as classification accuracy are therefore inadequate. Instead, we use measures based on precision, recall, and the intersection-over-union (IoU).

Beyond mAP_{50} and $\text{mAP}_{50:95}$ computed separately on each test set, we further use a multi-domain aggregate index to evaluate both performance and balance of the model across the entire deployment scenario.

Multi-domain harmonic mean ($\mathrm{HM}_{50}^{(3)}$). In real-world conditions, the model must maintain performance *simultaneously* on clear and foggy images in order to achieve the most general performance. The metric $\overline{\mathrm{mAP}}_{50}$ can overlook cases of overfitting to the training domains (where the training domains compensate for an untrained one), so the primary evaluation metric should directly reflect *multi-domain balance* rather than only the average performance. On this basis, we use the harmonic mean of mAP_{50} across the three domains: clear, synthetic fog and natural fog as the primary evaluation index, as in Equation 4.2.

$$\mathrm{HM}_{50}^{(3)} = \frac{3}{\frac{1}{\mathrm{mAP}_{50}^{\mathrm{clear}}} + \frac{1}{\mathrm{mAP}_{50}^{\mathrm{synthetic}}} + \frac{1}{\mathrm{mAP}_{50}^{\mathrm{real}}}}. \tag{4.2}$$

Unlike the arithmetic mean, the harmonic mean is sensitive to the smallest value: if a single domain has low performance, $\mathrm{HM}_{50}^{(3)}$ drops sharply even when the other two domains achieve high values. This approach is used to evaluate the balance between performance on seen and unseen domains [52].

In addition to the above metrics, the proposed method is also evaluated using metrics that depend on object size. AP_S, AP_M, and AP_L correspond to small, medium, and large objects, respectively. Specifically, following the COCO convention [53], objects with bounding-box area smaller than 32^2 pixels are considered small, between 32^2 and 96^2 pixels are medium, and larger than 96^2 pixels are large. These metrics reflect the detection capability of the model across different object-size scales and are particularly important for tasks involving many small objects or challenging viewing conditions such as fog.

4.4 Answer to RQs

4.4.1 RQ1 results: Comparison with DAOD methods

Table 4.4 summarizes the quantitative results answering RQ1, in which the proposed **FusionDA** (denoted Ours) is directly compared against two representative DAOD method families *DA-Detect* [4] and *ALDI++* [16] along with Full-finetune baselines on the same backbone. To ensure a fair comparison, the methods are organized into two detector groups: *one-stage* (the YOLO family) and *two-stage* (Faster R-CNN with the R50-FPN backbone). Each configuration is evaluated simultaneously on five datasets: Cityscapes, Foggy Cityscapes, and the three natural-fog test sets RTTS, FoggyDriving, and Fog-

Table 4.4: Comparison of DAOD methods on the Cityscapes/Foggy Cityscapes pair, together with zero-shot generalization evaluation on three natural-fog test sets (RTTS, FoggyDriving, FoggyZurich). All numbers are mAP_{50} (%); $\text{HM}_{50}^{(3)}$ is computed via Equation 4.2. Within each backbone group, the best $\text{HM}_{50}^{(3)}$ is in **bold**.

Backbone	**Method**	**mAP_{50}**					**$\text{HM}_{50}^{(3)}$ ↑**
		Cityscapes	**Foggy CS**	**RTTS**	**Foggy Drv.**	**Foggy Zur.**	
(I) One-stage detector							
YOLO26-n	Full-finetune	55.79	40.18	42.05	49.47	19.76	**43.00**
	Ours	50.14	40.44	50.97	48.40	19.35	42.89
YOLO26-s	Full-finetune	64.20	42.02	32.31	51.29	20.93	44.07
	Ours	61.31	54.03	57.18	59.01	30.10	**54.22**
YOLO26-m	Full-finetune	67.50	51.75	38.10	54.90	22.88	49.98
	Ours	64.29	57.58	42.20	55.27	27.33	**52.67**
YOLO26-l	Full-finetune	69.28	52.65	38.17	53.81	19.22	49.67
	Ours	65.05	59.53	47.23	59.95	29.10	**55.37**
YOLO26-x	Full-finetune	70.25	51.29	37.31	57.19	21.70	50.38
	Ours	66.14	60.07	44.55	59.59	25.41	**54.62**
YOLOv5-m	ALDI	54.96	43.82	43.57	44.46	26.17	44.58
	ALDI++ [16]	50.47	49.41	44.43	42.70	27.67	45.32
	Full-finetune	60.38	45.71	43.15	51.90	24.00	47.14
	Ours	57.26	47.40	47.46	53.12	24.52	**47.97**
(II) Two-stage detector							
R50-FPN	ALDI	50.63	41.11	45.47	45.87	26.35	43.12
	ALDI++ [16]	53.76	48.30	37.99	46.73	26.97	45.34
	DA-Detect base	54.89	35.84	42.23	41.83	19.10	39.89
	DA-Detect [4]	55.79	48.15	37.19	48.23	28.53	46.14
	Full-finetune	51.90	33.36	40.76	45.92	10.95	37.51
	Ours	53.06	48.01	46.01	48.45	26.24	**46.49**

gyZurich.

(1) FusionDA achieves the best balance across the three domains. FusionDA leads the $\text{HM}_{50}^{(3)}$ index in both detector groups (6 out of 7 backbones tested produce better results), confirming that it is the only method achieving an even balance across the clear, synthetic-fog, and natural-fog domains. In the *one-stage* group, FusionDA achieves substantial improvements: $\mathrm{HM}_{50}^{(3)}$ surpasses Full-finetune on the same backbones by +10.15 (s), +2.69 (m), +5.70 (l), and +4.24 (x) points, respectively. These designs do not suffer from the *catastrophic forgetting* typical of DAOD. For instance, the performance drop on the clear domain on YOLOv5m is −2.89 points, substantially smaller than that of ALDI++ (−4.49 points). The proposed method thus trades a small amount of accuracy on the clear domain for a substantial improvement across all other domains. This is consistent with the problem and the goal of our approach, namely that models degrade in performance under different conditions. In the *two-stage* group, FusionDA achieves $\mathrm{HM}_{50}^{(3)} = 46.49$, exceeding the second-best method DA-Detect (+0.35) and outperforming ALDI++ (+1.15), and in particular surpassing the corresponding Full-finetune configuration by +8.98 points.

The reason is the performance drop on the visually diverse RTTS dataset, as described in Table 4.1. DA-Detect and ALDI++ focus excessively on the synthetic-fog domain and overfit to driving scenes, which FusionDA avoids. This is precisely the value of physics-based domain translation compared to random augmentation (DA-Detect's RainMix) or standard data augmentation (ALDI++). There is an exception for the small model YOLO26-n, where balanced accuracy does not improve but instead drops slightly by −0.11 compared with Full-finetune. This is because training degrades on the clear domain without compensating gains on the other domains, failing to achieve the balance objective. This can be explained by the fact that the nano architecture is too small, with insufficient capacity to learn domain-invariant features, which leads to no improvement. In summary, FusionDA improves balance by learning domain-invariant features, as demonstrated in Section 4.4.2.

(2) FusionDA generalizes remarkably well to natural-fog domains. A core question for DAOD methods is whether they truly learn domain-invariant features or merely *overfit* to the synthetic-fog distribution of Foggy Cityscapes. To answer this, we perform zero-shot evaluation on three natural-fog test sets that are entirely absent from training. On all three sets, FusionDA improves substantially over the corresponding baseline. On *RTTS*, FusionDA-s reaches 57.18 versus the baseline 32.31, an increase of +24.87 points (+77.0% relative). This is the largest improvement in the entire study and surpasses both ALDI++ (37.99) and DA-Detect (37.19). On *FoggyZurich*, the most challenging set due to its low contrast, FusionDA-R50-FPN reaches 26.24 versus the Full-finetune baseline

Table 4.5: Per-size and per-class performance of **FusionDA** (Ours) against the Full-finetune baseline on *Foggy Cityscapes*. $\mathrm{AP}_S/\mathrm{AP}_M/\mathrm{AP}_L$ are $\mathrm{AP}_{50:95}$ within the three area ranges defined by COCO. $\mathrm{AP}^{\mathrm{Pers.}}$ and $\mathrm{AP}^{\mathrm{Car}}$ are per-class $\mathrm{AP}_{50:95}$. Within each Full-finetune/Ours pair, the better value is in **bold**. The last row summarizes the average improvement of FusionDA over the baseline (Δ = Ours − Full-finetune) across all backbones, indicating the magnitude of improvement along each evaluation axis.

Backbone	**Method**	**Overall**		**By size**			**By class**	
		$\mathrm{mAP}_{50:95}$	mAP_{50}	AP_S	AP_M	AP_L	$\mathrm{AP}^{\mathrm{Pers.}}$	$\mathrm{AP}^{\mathrm{Car}}$
YOLO26-s	Full-finetune	29.20	42.02	0.63	21.13	72.23	23.90	34.00
	Ours	**36.31**	**54.03**	**1.91**	**34.20**	**73.68**	**27.50**	**44.60**
YOLO26-m	Full-finetune	36.11	51.75	1.79	32.29	75.97	29.80	42.10
	Ours	**39.41**	**57.58**	**2.40**	**38.47**	**76.79**	**30.30**	**48.20**
YOLO26-l	Full-finetune	36.86	52.65	1.87	33.22	**77.88**	30.20	43.20
	Ours	**40.65**	**59.53**	**3.74**	**40.06**	77.31	**31.60**	**49.40**
YOLO26-x	Full-finetune	36.48	51.29	1.87	32.02	**77.95**	30.40	42.10
	Ours	**41.26**	**60.07**	**3.82**	**41.17**	77.89	**32.30**	**50.00**
R50-FPN	Full-finetune	20.05	33.36	0.17	10.38	59.26	16.46	23.64
	Ours	**27.37**	**48.01**	**1.11**	**21.72**	**61.49**	**19.37**	**35.37**
Average Δ (Ours − Full-finetune)				+1.33	+9.32	+0.77	+2.06	+8.51

of only 10.95, a relative improvement of +139.6%. Similarly, FusionDA-s improves FoggyZurich from 20.93 to 30.10 (+43.8% relative). On *FoggyDriving*, FusionDA-l reaches 59.95, exceeding DA-Detect (48.23) by 11.72 points. Notably, the relative improvements on the natural-fog sets *far exceed* the improvement on Foggy Cityscapes (+28.6% for YOLO26-s).

This is strong evidence that FusionDA truly learns fog-invariant representations rather than merely adapting to the synthetic-fog distribution. This finding aligns with the recent recommendation from the DAOD community [16] regarding the importance of including out-of-distribution test sets in addition to synthetic benchmarks when evaluating domain adaptation.

(3) To clarify how the proposed solution improves performance, the method is evaluated across different object sizes. The results indicate that **FusionDA improves most strongly on small and medium objects, which are the classic weakness under foggy conditions.** Table 4.5 and Table 4.6 provide a detailed analysis of performance

Table 4.6: Per-size and per-class performance of **FusionDA** (Ours) against the Full-finetune baseline on *RTTS* (natural fog, zero-shot). The reading convention follows Table 4.5.

Backbone	**Method**	**Overall**		**By size**			**By class**	
		$mAP_{50:95}$	mAP_{50}	AP_S	AP_M	AP_L	$AP^{Pers.}$	AP^{Car}
YOLO26-n	Full-finetune	24.96	42.05	9.53	31.68	49.36	25.30	24.10
	Ours	**31.02**	**50.97**	**11.10**	**37.59**	**52.30**	**30.70**	**31.10**
YOLO26-s	Full-finetune	19.15	32.31	7.81	24.15	44.77	21.10	16.40
	Ours	**35.56**	**57.18**	**17.81**	**41.23**	**55.19**	**34.70**	**36.50**
YOLO26-m	Full-finetune	22.98	38.10	8.82	30.37	38.40	22.70	22.40
	Ours	**25.84**	**42.20**	**12.33**	**33.19**	**40.29**	**25.30**	**26.10**
YOLO26-l	Full-finetune	22.79	38.17	9.56	29.52	44.44	21.60	23.10
	Ours	**28.46**	**47.23**	**13.53**	**36.28**	**44.68**	**28.10**	**28.80**
YOLO26-x	Full-finetune	22.72	37.31	9.32	29.50	40.68	22.80	22.10
	Ours	**27.26**	**44.55**	**12.85**	**35.38**	**42.28**	**26.40**	**27.80**
YOLOv5-m	Full-finetune	25.76	43.15	11.32	35.32	**38.15**	**26.00**	25.30
	Ours	**28.39**	**47.46**	**15.63**	**38.64**	37.74	25.70	**31.00**
R50-FPN	Full-finetune	22.66	40.76	10.49	28.80	37.09	22.26	23.06
	Ours	**25.38**	**46.01**	**12.75**	**33.14**	**38.31**	**24.58**	**26.19**
Average Δ (Ours − Full-finetune)				+4.16	+6.59	+2.41	+4.82	+7.29

by object size (small, medium, large) and by class (Person, Car) on the two foggy sets. The improvement of FusionDA grows with smaller-to-medium object sizes on both domains: on Foggy Cityscapes, AP_M increases by an average of +9.32 points and AP_S by +1.33 points (significant given that the baseline on small objects is very low). Meanwhile, AP_L rises by only +0.77 points because the baseline is already close to saturation in the 72–78 range. The generalization of FusionDA to natural fog is particularly strong in the small-object segment: the average ΔAP_S on RTTS reaches +4.16 points, larger than the improvement on Foggy Cityscapes (+1.33), and especially with the YOLO26-s configuration AP_S rises by +128%. The per-class analysis shows that *Car* is consistently improved more than *Person* on both domains (+8.51 vs. +2.06 on Foggy CS; +7.29 vs. +4.82 on RTTS), reflecting the fact that pedestrians in fog are both small in size and easily confused with the background.

Thus, distant pedestrians and partially occluded vehicles are simultaneously the hardest objects to detect. The strong improvement of FusionDA on AP_S and AP_M, particularly pronounced on natural-fog data, indicates that the proposed method *addresses the core* *weakness* of detectors under adverse weather, rather than merely raising the average quality by improving large objects that are already easy to detect even at baseline.

To further support the small-object improvement of the method, a qualitative analysis on the small subset of Foggy Cityscapes (Figure 4.3) shows that the number of true positives on small objects consistently improves from Baseline to FusionDA. For example, in row 1, Baseline detects no small object (TP= 0, FN= 9), while the full FusionDA reaches 6; a similar trend appears in rows 2 and 3, especially in dense-fog regions near the image center.

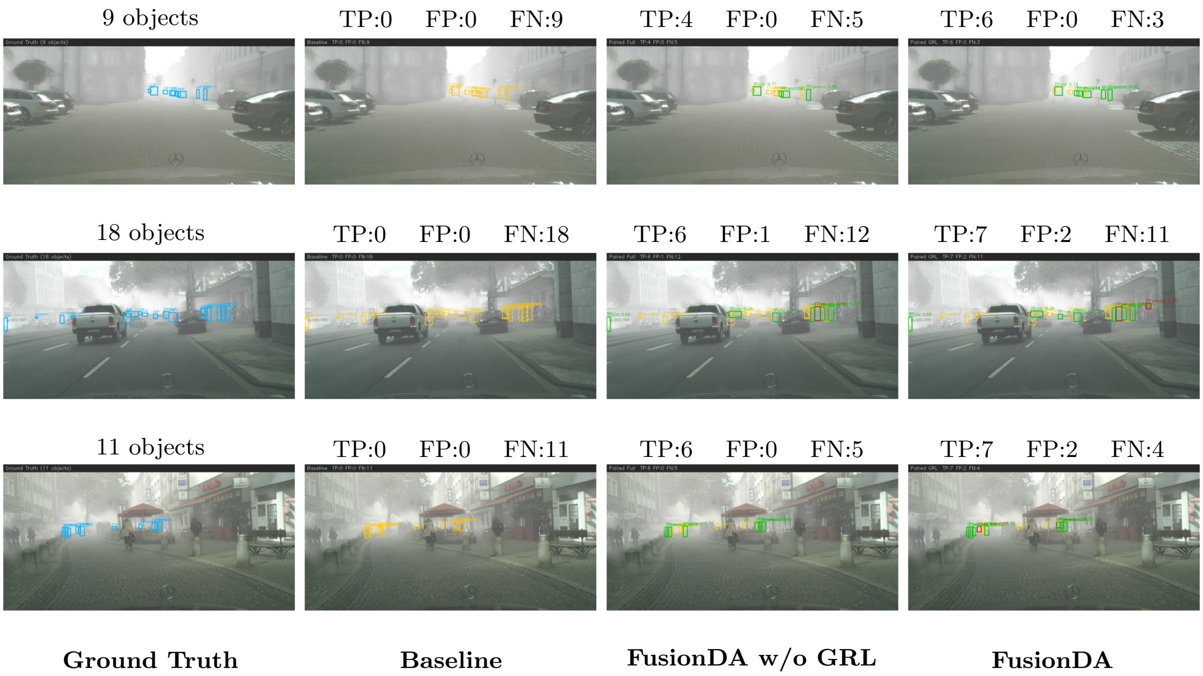

Figure 4.3: Comparison of detection results on the small-object subset of the Foggy Cityscapes validation set. Blue box: ground truth; green box: true positive; red box: false positive; dashed yellow box: false negative.

This qualitative observation is consistent with the quantitative findings reported in Tables 4.5 and 4.6: the improvement of FusionDA grows with smaller object sizes, while the accuracy on large objects rises only marginally because the baseline is already near saturation.

4.4.2 RQ2 results: Component contribution analysis

To answer RQ2, this thesis performs an ablation study by progressively integrating each component of FusionDA, thereby evaluating the individual contribution of each module.

The investigated configurations are described in detail in Table 4.7.

The configurations are designed for systematic ablation. Configuration (a) Standard fine-tuning baseline serves as the lower bound on performance. The two configurations (b) Distillation only and (c) Augment only represent the two independent branches of FusionDA: (b) activates only the Mean Teacher framework to distill knowledge on unlabeled foggy data, while (c) activates only the supervised training branch on the pseudo-foggy images, thereby allowing us to quantify the role of each mechanism in isolation before combining them. Configuration (d) Augment Distillation merges the two branches into a single cross-domain distillation model, clarifying the complementarity or conflict between MT and $\mathbf{I}^{sf}$. Finally, (e) FusionDA w/o GRL adds the feature-consistency loss $\mathcal{L}_{con}$ between the pair $(\mathbf{I}^{s}, \mathbf{I}^{sf})$, and (f) *FusionDA* completes the architecture with the Gradient Reversal Layer module.

Table 4.7: Description of experimental configurations in the FusionDA ablation study. The mark ✓indicates an activated component. **MT** is the Mean Teacher framework; $\mathbf{I}^{sf}$ is the supervised training branch on pseudo-foggy images; $\mathcal{L}_{con}$ is the feature-consistency loss; GRL is the Gradient Reversal Layer module. Configuration **(f)** is the full FusionDA.

Configuration	**MT**	$\mathbf{I}^{sf}$	$\mathcal{L}_{con}$	**GRL**	**Notes**
(a) Baseline					clear-image fine-tuning only
(b) Distillation only	✓				self-distillation only
(c) Augment only		✓			pseudo-foggy images only
(d) Augment Distillation	✓	✓			combined distillation
(e) FusionDA w/o GRL	✓	✓	✓		full, without GRL
(f) **FusionDA**	✓	✓	✓	✓	**proposed method**

All configurations use the same YOLO26-small backbone and are trained on the labeled training set of Cityscapes and the unlabeled training set of Foggy Cityscapes. Evaluation is conducted simultaneously on three domains: clear, synthetic fog, and natural fog. The quantitative results are reported in Table 4.8.

The components complement each other to achieve multi-domain balance. Table 4.8 shows that the complete FusionDA (configuration (f)) achieves the highest balanced-accuracy index $\mathrm{HM}_{50}^{(3)} = 54.22$, exceeding the Baseline (configuration (a)) by +10.15 points, a +23.0% relative improvement. More importantly, FusionDA leads mAP_{50} on the natural-fog domain, exceeding the Baseline (34.84) by +13.92 points. This result

Table 4.8: Ablation study of the FusionDA components, evaluated simultaneously on Cityscapes, Foggy Cityscapes, and the natural-fog domain. $\mathrm{HM}_{50}^{(3)}$ is computed via Equation 4.2. The best result per column is in **bold**, the second best is underlined.

Configuration	**mAP$_{50}$**			**HM$_{50}^{(3)}$** $\uparrow$
	Cityscapes	**Foggy CS**	**Real Fog**	
(a) Baseline	**64.20**	42.02	34.84	44.07
(b) Distillation only	60.37	44.36	46.58	49.52
(c) Augment only	<u>63.17</u>	**60.68**	32.31	47.42
(d) Augment Distillation	59.39	50.44	<u>47.51</u>	51.98
(e) FusionDA w/o GRL	60.85	53.88	46.67	<u>53.17</u>
(f) **FusionDA**	61.31	<u>54.03</u>	**48.76**	**54.22**

is particularly meaningful in practice because the Real Fog domain does not appear in training and represents the actual deployment scenario. Notably, configuration (c) $+\mathbf{I}^{sf}$ alone achieves the highest $\mathrm{mAP}_{50}^{\mathrm{tgt}}$ (60.68), even surpassing the full FusionDA, but at the same time falls to the lowest score on Real Fog (32.31, even below Baseline). This is a clear sign of *overfitting to the synthetic-fog distribution.* The full FusionDA, in contrast, trades -6.65 points on the synthetic-fog domain to gain $+16.45$ points on the natural-fog domain, while maintaining high performance on the clear domain (only -2.89 points below Baseline). As a result, the $\mathrm{HM}_{50}^{(3)}$ index increases monotonically along the chain of progressive integration (a)$\mapsto$(b)$\mapsto$(d)$\mapsto$(e)$\mapsto$(f). This confirms that the components are *complementary* rather than overlapping, and that each addition improves multi-domain balance rather than trading one domain for another.

This is strong quantitative evidence that the FusionDA design prioritizes *general domain invariance* over peak performance on any specific dataset. Figure 4.4 compares predictions of three configurations on representative images from the Foggy Cityscapes validation set. The number of missed objects (false negatives, dashed yellow boxes) drops sharply when moving from Baseline to FusionDA w/o GRL, while FusionDA further recovers objects that FusionDA w/o GRL still misses, particularly in scenes with dense fog or backlit lighting.

Quantitative contribution of each component. The quantitative analysis shows that each component plays a *distinct but complementary* role in FusionDA:

- **Mean Teacher alone, (b) vs. (a):** $+5.45$ $\mathrm{HM}_{50}^{(3)}$. The self-distillation framework

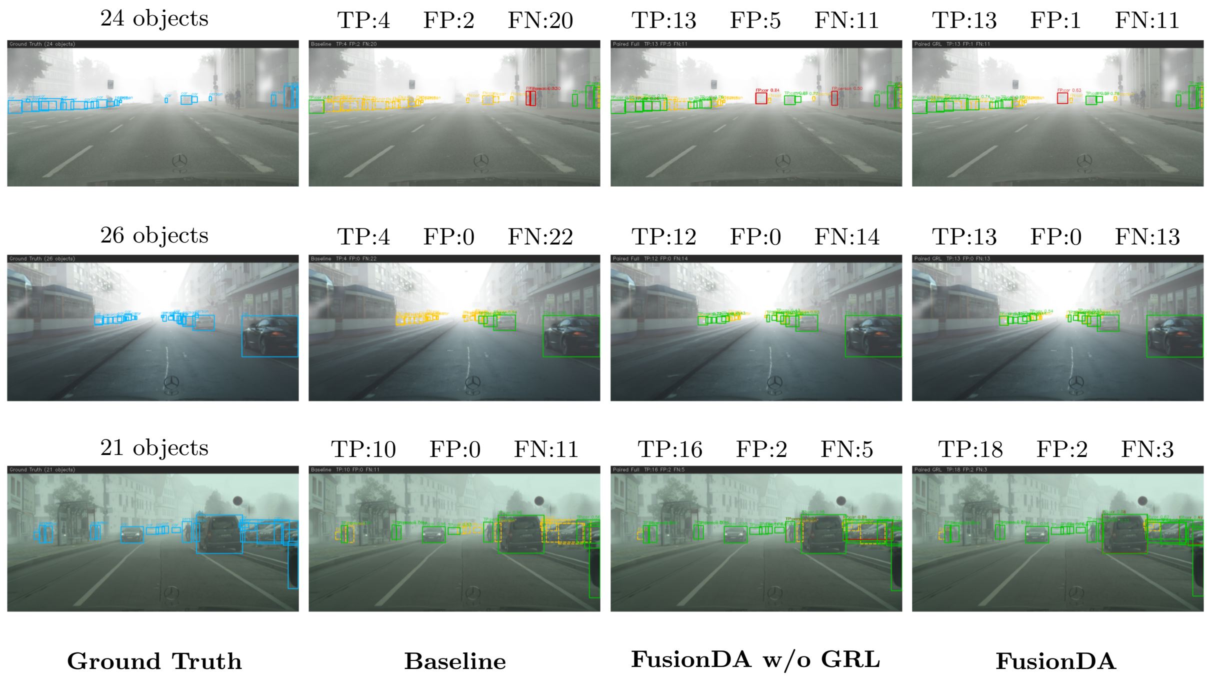

Figure 4.4: Visual comparison of detection results on the Foggy Cityscapes domain. Blue box: ground truth; green box: true positive; red box: false positive; dashed yellow box: false negative.

exploits foggy-domain data and boosts mAP_{50} on Real Fog by +11.74 points (34.84 $\rightarrow$ 46.58). This provides the foundation for zero-shot generalization to natural fog. However, MT alone yields only a modest improvement on the synthetic-fog domain (+2.34 $\text{mAP}_{50}^{\text{tgt}}$).

- **Augmented training alone, (c) vs. (a):** +18.66 $\text{mAP}_{50}^{\text{synthetic}}$ (42.02 $\rightarrow$ 60.68, *the highest in the entire table*), but −2.53 $\text{mAP}_{50}^{\text{real}}$ (34.84 $\rightarrow$ 32.31, *below Baseline*). The supervised branch on pseudo-foggy images directly optimizes the Foggy Cityscapes distribution, leading to *overfitting to synthetic fog* and weakening generalization. This observation motivates the need to *combine* $\mathbf{I}^{sf}$ with MT in configuration (d) rather than using it in isolation.

- **MT + $\mathbf{I}^{sf}$, (d) vs. (c):** +15.20 $\text{mAP}_{50}^{\text{real}}$ (32.31 $\rightarrow$ 47.51). Although $\text{mAP}_{50}^{\text{tgt}}$ drops (60.68 $\rightarrow$ 50.44), $\text{HM}_{50}^{(3)}$ still increases by +4.56 points (47.42 $\rightarrow$ 51.98), confirming that multi-domain balance matters more than peak performance on a single domain.

- **Feature-consistency loss $\mathcal{L}_{con}$, (e) vs. (d):** +1.19 $\text{HM}_{50}^{(3)}$ (51.98 $\rightarrow$ 53.17). The cosine constraint between $\phi_s(\mathbf{I}^s)$ and $\phi_s(\mathbf{I}^{sf})$ recovers +1.46 points on the clear domain (59.39 $\rightarrow$ 60.85) and improves the target domain by +3.44 points (50.44 $\rightarrow$ 53.88), at a small cost of -0.84 points on Real Fog. The overall balance still improves thanks to the constraint enforcing style-invariant features on the pair $(\mathbf{I}^s, \mathbf{I}^{sf})$.

- **GRL module, (f) vs. (e):** $+1.05$ $\mathrm{HM}_{50}^{(3)}$ ($53.17 \to 54.22$). Although this is the smallest quantitative improvement among the components, GRL raises $\mathrm{mAP}_{50}^{\mathrm{real}}$ by $+2.09$ points ($46.67 \to 48.76$), reaching the highest Real Fog value in the entire ablation study. The feature-level adversarial mechanism forces the backbone to learn truly domain-invariant representations. The qualitative role of the GRL is analyzed further below.

Analysis of the domain gap in feature space. To verify the hypothesis that the proposed components truly narrow the domain gap (rather than merely improving the final performance), we extract features from the last layer of the backbone on 175 random images from each domain and compute the *Maximum Mean Discrepancy* (MMD) [54] with a Gaussian kernel, while also projecting them onto a two-dimensional space using UMAP [55] for visualization. The MMD between two domains S and T is computed via Equation 4.3, the biased estimator. The smaller the MMD, the smaller the feature distance between the two domains, indicating that the domains are difficult to distinguish.

$$\widehat{\mathrm{MMD}}^2(S,T) = \frac{1}{m(m-1)}\sum_{i\neq i'} k(s_i,\, s_{i'}) - \frac{2}{mn}\sum_{i,j} k(s_i,\, t_j) + \frac{1}{n(n-1)}\sum_{j\neq j'} k(t_j,\, t_{j'}) \tag{4.3}$$

where S and T represent the two domains whose distance is being computed, $\{s_i\}_{i=1}^m$ and $\{t_j\}_{j=1}^n$ are their respective feature vectors, with $m = n = 175$. The kernel $k(\cdot,\cdot)$ is a linear combination of multiple Gaussian kernels with different σ_ℓ, as in Equation 4.4.

$$k(x,\, y) = \sum_{\ell} \exp\left(-\frac{\|x-y\|^2}{2\,\sigma_\ell^2}\right) \tag{4.4}$$

The results are reported in Figure 4.5 and Table 4.9.

Figure 4.5 shows that the Baseline representation clearly separates clear-domain features (blue) and foggy-domain features (orange) into two nearly disjoint clusters. After applying FusionDA w/o GRL, the two distributions become heavily mixed and MMD drops to ≈ 0, confirming that the domain translation mechanism combined with $\mathcal{L}_{con}$ has almost entirely removed the domain gap at the statistical level. Although both FusionDA w/o GRL and FusionDA achieve MMD $= 0$, the UMAP plot reveals a difference in *spatial uniformity*: FusionDA w/o GRL still exhibits noticeably sparse regions, indicating that

Table 4.9: MMD distance between clear-domain and foggy-domain features, measured on the backbones of three representative configurations. The lower the distance, the closer the features and the harder they are to distinguish.

Configuration	**MMD(clear $\leftrightarrow$ foggy) $\downarrow$**
(a) Baseline	2.55×10^{-3}
(d) FusionDA w/o GRL	≈ 0
(e) **FusionDA**	≈ 0

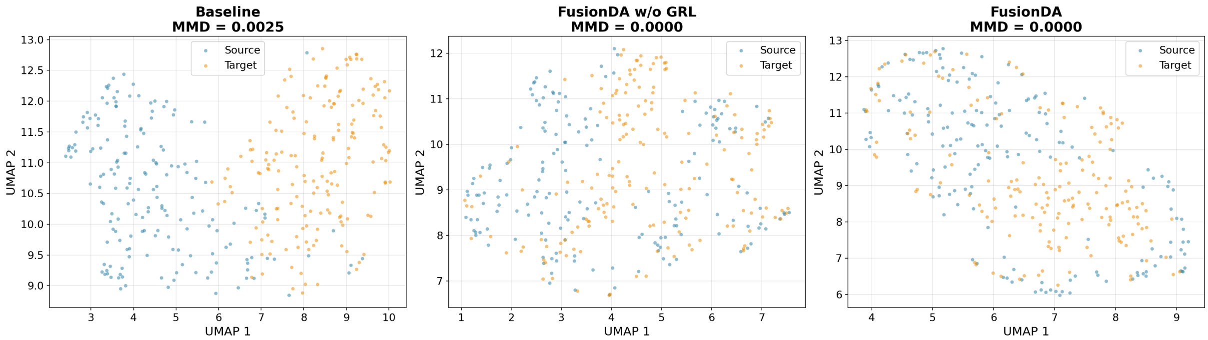

Figure 4.5: UMAP visualization of the backbone feature space on 175 clear and 175 foggy samples. **Left:** the Baseline shows a clear separation between the two domains along the UMAP-1 axis. **Middle:** FusionDA w/o GRL - the two distributions are substantially mixed and MMD drops to approximately 0. **Right:** FusionDA (full) - the two distributions become more uniformly arranged in terms of local structure, even though the MMD is already saturated.

the feature space is unevenly distributed even though the global statistics have converged. In contrast, the full FusionDA fills the UMAP space more evenly, leaving no large empty regions, showing that the GRL promotes *locally consistent* domain alignment across the entire feature space, not only at the global average level that MMD measures. This observation suggests that the GRL continues to refine the local structure of the feature space toward domain invariance, *consistent with the quantitative findings* in Table 4.8: the GRL recovers +2.47 points of mAP_{50} on Real Fog over configuration (d).

The role of GRL through attention analysis The UMAP analysis above reveals that the GRL does contribute to the local quality of the feature space, a dimension that MMD and mAP do not measure directly. To understand the specific mechanism behind this local difference, we look deeper into the backbone by visualizing the attention maps, rather than only observing the final detection outputs. YOLO26 ends its backbone with a *C2PSA* block (Cross-Stage Partial with Partial Self-Attention) [56], which is the final layer that aggregates all spatial information before passing it to the neck and head. The core of this block is a *multi-head self-attention* mechanism. The position of C2PSA in the architecture makes it an ideal place to observe: if the backbone has truly learned domain-invariant representations, the attention at this layer must reflect this property through how each token attends to the semantic regions of the image, independent of the domain style.

In multi-head self-attention, each *query token* is a feature vector at a position on the feature map, acting as the "query". The model then computes the similarity between q_i and all remaining *key tokens* k_j according to Equation 4.4.2.

$$a_{ij} = \mathrm{softmax}\left(\frac{q_i^\top k_j}{\sqrt{d_k}}\right),$$

The result is a probability distribution $\{a_{ij}\}_{j=1}^{N}$ over the entire feature space, called the attention map of token i. A large coefficient a_{ij} means that token i is "attending strongly" to token j, i.e., the model considers the two regions to be strongly related semantically. The value d_k is the dimension of the key vector and serves as a normalization factor to stabilize the gradient.

To choose query tokens with semantic meaning, the visualization places the query at the center of each detected bounding box. This ensures that the token is querying a region containing an object, not a random background area. The attention map is averaged across heads and reshaped to $H_{\mathrm{feat}} \times W_{\mathrm{feat}}$, then bilinearly upsampled to the original image size and overlaid on the input image as a *heat map* (red: high attention; blue: low attention).

Figure 4.6 shows clearly different attention patterns under the same "car" query:

- **Baseline:** attention is widely scattered and does not converge clearly onto any object. When the model encounters a foggy image, these features become noisy and lose reliable localization.

- **FusionDA w/o GRL:** attention begins to localize object regions (especially clearly near "car" queries), but secondary peaks still appear scattered on unrelated background regions and the spread to other same-class objects is still weak. This indicates that the Mean Teacher and the pseudo-foggy images have helped the model encounter

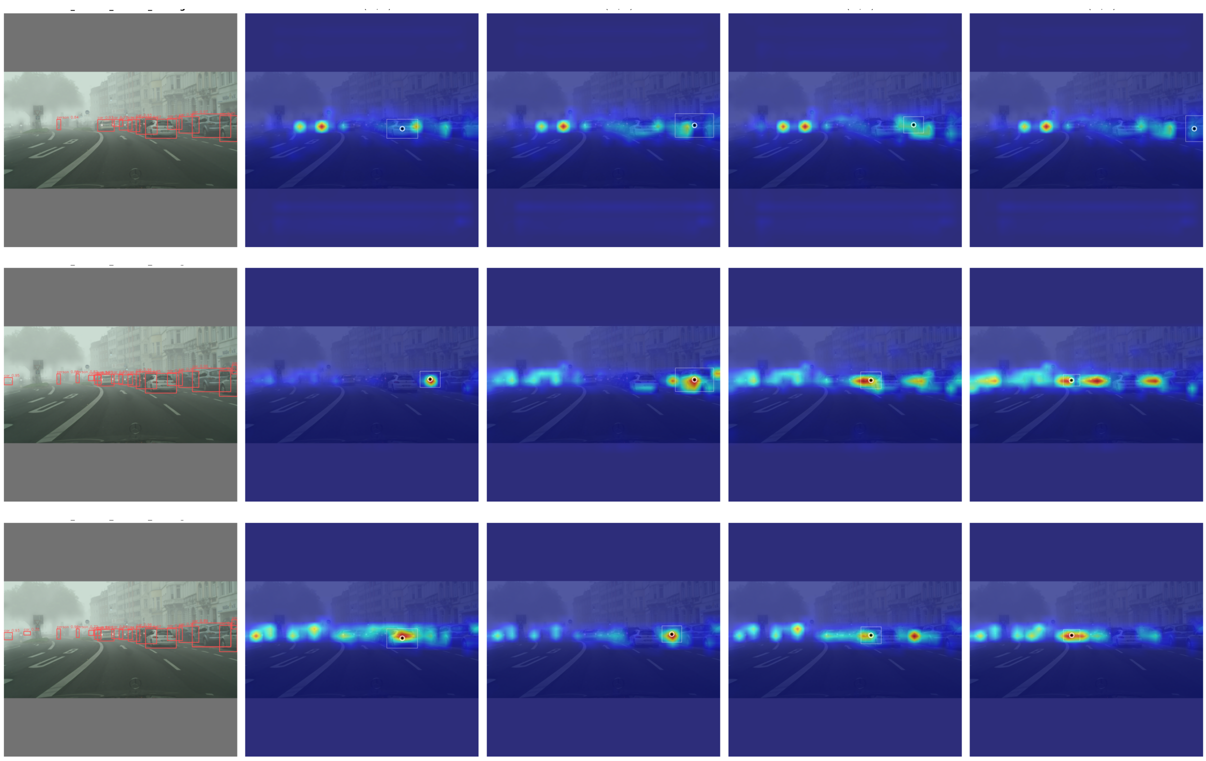

Figure 4.6: Comparison of C2PSA attention maps for the same input image (`aachen_000131_00001_leftImg8bit`) across three configurations. **Top row:** Baseline (source-only). **Middle row:** FusionDA w/o GRL. **Bottom row:** FusionDA (full). Each column corresponds to a query token placed at the location of a car. The brighter the region, the more the model attends to it.

the foggy distribution and learn more stable representations, but they have not yet completely eliminated the influence of domain-style features.

- **FusionDA (full):** attention is sharply focused on the query object and structurally spreads to other same-class objects at different positions in the image (cars parked along both sides of the street). This is the desired behavior of a domain-invariant detector: the model relies on object-level semantic features (shape, wheel structure, body outline) rather than domain-style features (fog density, background contrast) to make predictions.

Thus, the GRL acts as a *local refinement of representation quality*. This is a clear example of how the qualitative quality of features can exceed what common quantitative metrics can capture.

Table 4.10: Performance of **FusionDA** (Ours) against the Full-finetune baseline on the WIDERFACE-easy dataset. $\mathrm{mAP}_{50}^{\mathrm{clear}}$ and $\mathrm{mAP}_{50}^{\mathrm{foggy}}$ are computed on the clear and foggy domains; HM_{50} is the harmonic mean of the two domains. $\mathrm{AP}_S/\mathrm{AP}_M/\mathrm{AP}_L$ are computed on the foggy domain following the COCO convention. Within each Full-finetune/Ours pair, the better value is in **bold**. The last row summarizes the average improvement ($\Delta =$ Ours $-$ Full-finetune) across backbones.

Backbone	**Method**	**Overall**			**By size (Tgt)**		
		$\mathrm{mAP}_{50}^{\mathrm{clear}}$	$\mathrm{mAP}_{50}^{\mathrm{foggy}}$	HM_{50}	AP_S	AP_M	AP_L
YOLO26-n	Full-finetune	93.50	**93.87**	**93.68**	17.03	**63.78**	**71.69**
	Ours	**93.92**	93.05	93.48	**18.54**	62.79	71.10
YOLO26-s	Full-finetune	94.12	**94.07**	94.09	14.30	**65.39**	**72.06**
	Ours	**94.95**	93.98	**94.46**	**23.72**	64.73	71.63
YOLO26-m	Full-finetune	94.19	94.24	94.21	13.44	65.75	71.95
	Ours	**95.40**	**95.01**	**95.20**	**27.02**	**65.89**	**72.12**
YOLO26-l	Full-finetune	94.77	94.68	94.72	17.43	66.21	**72.45**
	Ours	**95.69**	**95.18**	**95.43**	**29.53**	**66.32**	72.23
YOLO26-x	Full-finetune	94.58	94.50	94.54	18.95	65.98	72.24
	Ours	**95.57**	**94.63**	**95.10**	**27.00**	**66.46**	72.24
Average Δ (Ours $-$ Full-finetune)					+8.93	−0.18	−0.21

4.4.3 RQ3 results: Generalization to different tasks and corruption types

To strengthen the generality of FusionDA beyond the Cityscapes/Foggy Cityscapes benchmark, this thesis conducts a supplementary experiment on a different scenario. This is a single-class detection task (face). Specifically, the WIDERFACE-easy dataset [29] serves as the clear domain, while the foggy domain is generated online by the `fog` transformation of `imagecorruptions` at random levels on the same set of images. The experimental results are reported in Table 4.10. This is a substantially *easier* scenario than Cityscapes (the baseline already reaches $\sim 94\%$), so this experiment does not aim to demonstrate peak improvement but rather serves as a *generalization sanity check*: verifying that the mechanisms of FusionDA remain effective when faced with a different label structure and corruption type.

FusionDA maintains its improvement trend even on labels and corruption types outside its original design. FusionDA improves AP_S by an average of +8.93 absolute points across all backbones. This observation *exactly repeats* the strong small-object improvement trend observed on Cityscapes/Foggy Cityscapes (Insight (3), Section 4.4.1). HM_{50} rises only slightly by +0.47 points despite the baseline being already in the saturation regime ($\sim 94\%$), directly reflecting the saturation effect on easy data.

This finding shows that the effectiveness of FusionDA does not depend on the specific design of the Cityscapes/Foggy Cityscapes benchmark. FusionDA continues to strongly improve AP_S and maintain the HM_{50} balance, reinforcing the hypothesis that the $\mathbf{I}^{sf}$ synthesis mechanism combined with the feature-consistency constraint $\mathcal{L}_{con}$ truly teaches the backbone to learn robust representations for small objects. This is a capability that transfers to any detection task and is not tied to a specific label type or corruption source. The fact that $\mathrm{AP}_M/\mathrm{AP}_L$ and $\mathrm{mAP}_{50}^{\text{foggy}}$ remain nearly unchanged shows that FusionDA focuses its improvements on the hardest segment (small objects) rather than distributing them uniformly.

4.5 Threat to validity

To ensure that conclusions about the performance of FusionDA relative to the comparison methods are objective and reproducible, the experimental design follows the principle of fairness.

Fairness of data. All comparison methods and the proposed method are trained on the same data: the labeled training set of Cityscapes and the unlabeled training set of Foggy Cityscapes. Evaluation is also conducted on the same test splits: the Cityscapes validation set, the Foggy Cityscapes validation set, and the three natural-fog sets RTTS, FoggyDriving, and FoggyZurich-test, with the same number of images and the same splits as in Table 4.1. The diversity of the datasets also ensures generality and avoids bias toward any single domain. All methods are evaluated on the same two object classes *person* and *car*, with the same input resolution. The only asymmetry lies in internal augmented data: DA-Detect [4] uses RainMix as an auxiliary domain, ALDI++ [16] applies standard strong augmentations, and FusionDA generates cross-domain image pairs $(\mathbf{I}^{sf}, \mathbf{I}^{tf})$ from the physical scattering model which is itself a strength of the method. However, none of these three augmentation methods *uses external data* beyond the original training set, so they remain within the same standard-data scope.

Fairness of the data generation models. The two models used in the data generation phase: DepthAnythingV2 [49] and AODNet [5]. They are both open-source models with publicly available checkpoints that are not trained or fine-tuned on any of the experimental datasets. Their role is entirely *outside the training/evaluation loop*: the images $\mathbf{I}^{sf}$ and $\mathbf{I}^{tf}$ are generated once before training begins, and no comparison method has access to this augmented image set. This is also not an asymmetric advantage in terms of *external data*, since DAOD methods all augment internal data in their own way, and no method gains additional labeled data from external sources. Ultimately, the role of these data-generation models is that of a data preprocessor, similar to common image-augmentation filters. The true effectiveness of FusionDA comes from the training strategy rather than the quality of the synthesized images, as confirmed by the component ablation study in Section 4.4.2.

Fairness of backbone usage. To rule out the influence of architecture on the evaluation of the DA algorithm, FusionDA is validated across *diverse backbones*, including those used by the original baselines: *R50-FPN* (shared with DA-Detect and ALDI++), YOLOv5-m (shared with ALDI++), and the YOLO26 family from *nano* to *x-large* to investigate the scalability with respect to model capacity. All methods use the same pretrained weights from ImageNet/COCO for the corresponding backbone. On the R50-FPN backbone, the comparison becomes *absolutely symmetric* with respect to parameter capacity and the pre-training pipeline, thereby allowing performance differences to be attributed directly to the domain-adaptation algorithm.

Fairness of evaluation All methods are evaluated by *the same procedure* based on the standard pycocotools library to compute mAP_{50}, $\mathrm{mAP}_{50:95}$, $\mathrm{AP}_{S/M/L}$, and per-class AP, ensuring no discrepancy due to the evaluation library implementation. The harmonic-mean index $\mathrm{HM}_{50}^{(3)}$ is computed by the same Equation 4.2 for every method. More importantly, testing on the natural-fog sets is *entirely unbiased*, because no method is fine-tuned directly on these three sets during training. Therefore, every method faces the same *domain gap* when evaluated out-of-distribution, ensuring that the numbers on the three Real Fog sets faithfully reflect the generalization capability of each algorithm. Finally, to ensure independent reproducibility, the full source code, model checkpoints, and raw result files are publicly released so the research community can verify the numbers reported in this thesis.

4.6 Limitations and future work

Although FusionDA achieves substantial improvements, several observations should be noted for future research. *First*, the marginal quantitative contribution of GRL on Foggy Cityscapes suggests that better coordination between the GRL's adversarial coefficient λ and modifications to the GRL module architecture could make a difference. *Second*, the *person* class is consistently improved less than the *car* class and small objects remain difficult to detect across all backbones and all foggy sets (Tables 4.5 and 4.6). This reflects the fact that pedestrians in fog are both small in size and easily confused with the background, calling for finer-grained instance-level feature-exploitation mechanisms in future work. *Third*, FusionDA is currently validated on only one type of weather corruption (fog); extending it to other adverse-weather conditions (rain, snow, low light) and to more diverse domain pairs (such as Cityscapes/BDD100K or KITTI/Foggy KITTI) is a natural future direction to confirm the generality of the proposed design.

Conclusion

Object detection under foggy conditions is a problem of significant practical importance for autonomous-driving systems and outdoor intelligent surveillance, yet modern detectors such as YOLO and Faster R-CNN suffer severe performance degradation when deployed outside the training domain due to light scattering, reduced contrast, and the distribution shift between the labeled clear domain and the unlabeled foggy domain. To address these challenges simultaneously, this thesis proposes **FusionDA**, a semi-supervised domain-adaptation method that integrates three complementary mechanisms end-to-end at different alignment levels: (i) generating cross-domain image pairs $\mathbf{I}^{sf}$ and $\mathbf{I}^{tf}$ based on the atmospheric scattering model combined with the DepthAnythingV2 depth estimator and the AODNet dehazing network to narrow the domain gap at the input-image level; (ii) the Mean Teacher framework with knowledge distillation to exploit unlabeled foggy-domain data through high-quality pseudo-labels; and (iii) the Gradient Reversal Layer (GRL) module combined with the feature-consistency loss $\mathcal{L}_{con}$ to force the backbone to learn domain-invariant representations at the feature level.

Results. Comprehensive experiments on the standard Cityscapes and Foggy Cityscapes benchmark, compared against two representative DAOD methods (DA-Detect and ALDI++) on both *two-stage* (Faster R-CNN) and *one-stage* (YOLOv5, YOLO26 from *nano* to *x-large*) architectures, show that FusionDA leads on most quantitative metrics. Specifically, on the YOLO26 family, FusionDA achieves mAP_{50} on the foggy domain ranging from 54.03 (YOLO26-s) to 60.07 (YOLO26-x), with the harmonic-mean index $\text{HM}_{50}^{(3)}$ remaining in the range 54.22–55.37 across the YOLO26-{s,m,l,x} variants (peaking at YOLO26-l), surpassing the strongest existing DAOD baseline ($\text{HM}_{50}^{(3)} = 46.14$ of DA-Detect R50-FPN) by +8.08 to +9.23 absolute points, corresponding to relative improvements of +17.5% to +20.0%. More notably, under *zero-shot* evaluation on natural-fog sets that are entirely absent from training, FusionDA improves AP_{50} by +24.87 points (+77.0% relative) over the source-only baseline; this relative gain *far exceeds* the gain on the synthetic Foggy Cityscapes domain (+28.6%), confirming that FusionDA truly learns representations invariant to the general phenomenon of "fog" rather than merely overfitting to the synthetic distribution. To further verify the generality of FusionDA beyond the Cityscapes/Foggy Cityscapes benchmark, this thesis conducts a supplementary experiment (RQ3) on the WIDERFACE-easy face-detection dataset [29]. Compared with the *Full-finetune* baseline on the foggy domain, FusionDA improves AP_S by +8.93 points on average uniformly across all five YOLO26 variants, while maintaining cross-domain balance with an HM_{50} change of only −0.21 points. This result shows that the advantage of FusionDA transfers to other practical configurations, reinforcing the hypothesis that the method learns representations invariant to image-quality degradation at a general level.

Main Contributions. This thesis makes three main contributions to the research community: (i) we propose the **FusionDA** architecture with a multi-level complementary component design that simultaneously narrows the domain gap at the image and feature levels; (ii) we establish a comprehensive evaluation protocol on the standard Cityscapes/Foggy Cityscapes pair together with zero-shot evaluation on natural-fog data, along with a detailed ablation analysis of each component and a qualitative analysis via MMD/UMAP/attention; and (iii) we publicly release the full source code, model checkpoints, and generated datasets to ensure reproducibility and support future research on object detection under adverse weather.

Limitations and Future Work. Although FusionDA achieves substantial improvements, several limitations remain to be addressed in future research. *First*, the quantitative contribution of GRL on Foggy Cityscapes is marginal (+0.15 points of AP_{50}), suggesting that the $\mathbf{I}^{sf}$ and $\mathcal{L}_{con}$ components have absorbed most of the benefit from feature alignment; better coordination between the GRL's λ schedule and the weight $w_{sf}(t)$ is a worthwhile direction for improvement. *Second*, $\mathcal{L}_{con}$ introduces a slight fluctuation on Real Fog (a −0.84-point drop in mAP_{50} at step (d)→(e) before GRL recovers and pushes it up by +2.09 points at step (f)), suggesting that the feature-level consistency mechanism could be further refined by dynamically adjusting the weight λ_{con} along the training progress, or by constraining consistency only on a semantically selected feature subset. *Third*, the YOLO26-nano variant shows limited improvement because its parameter capacity is insufficient to simultaneously train multi-domain supervised branches and maintain a stable clear-domain representation; applying knowledge-distillation techniques from large to small models is a practically significant next direction for edge devices. In the future, FusionDA can be naturally extended to other adverse weather conditions (rain, snow, low light), can integrate more state-of-the-art depth-estimation and dehazing networks, or can explore finer-grained cross-domain attention mechanisms to further push the boundaries of object detection under real-world deployment.

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